CHEMISTRY OF ADVANCED ENERGY STORAGE DEVICES FOR E-MOBILITY(18G6E14)

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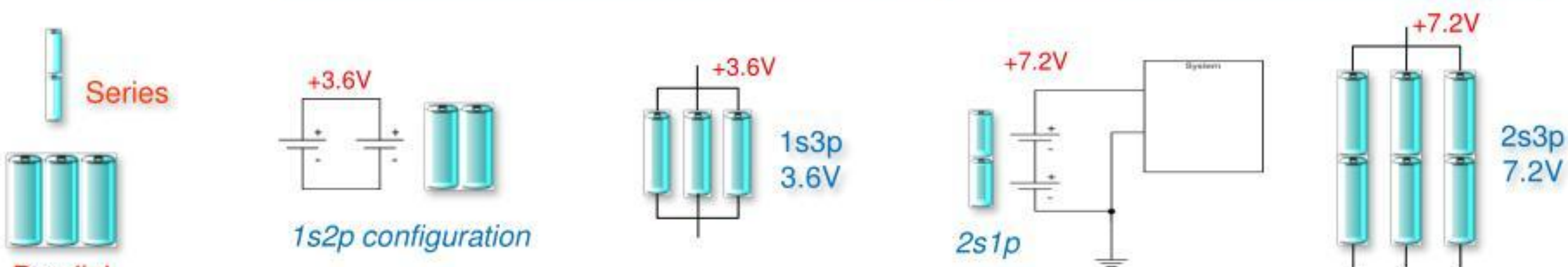
What is Battery Management?

- **Battery management circuits**
  - *Control charge flow into battery*
  - *Prevent abuse conditions*
  - *Monitor critical parameters*
  - *Communicate information*

- **Systems require battery management to extend run-times, maximize safety, and maximize battery-life**

- **Degrees of implementation vary, but sophisticated battery management consists of <u>three</u> components : Battery Charge Mgmt, Battery Gauge / Mgmt, and Protection.**

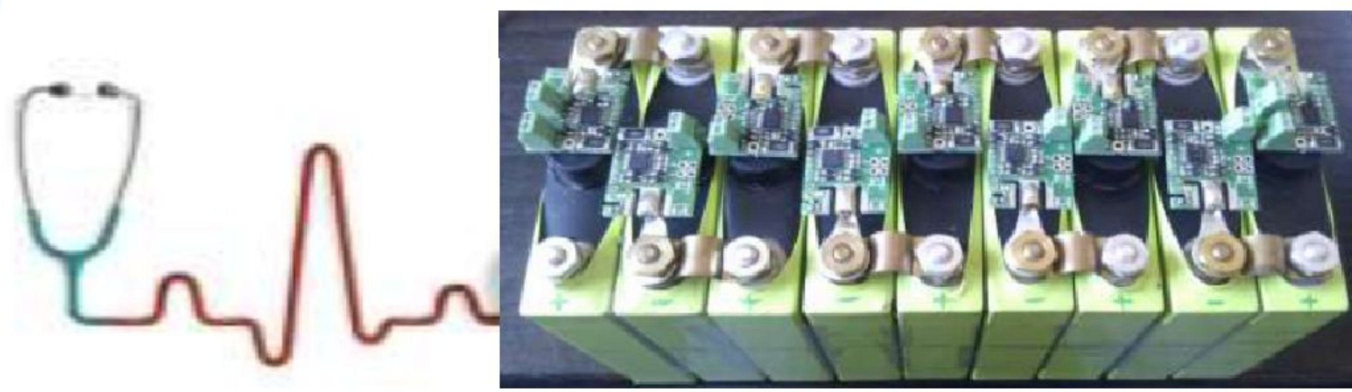
Pack Configurations



Battery Management system: Aim

BMS system has for objective:

- Protection and prevention of the system from damage
- Increase of battery life
- Maintenance of the battery system in accurate and reliable state
- BMS = Battery Doctor



Tesla battery module https://www.youtube.com/watch?v=bNd-yJtRPhk

BMS: Fuel Gauging

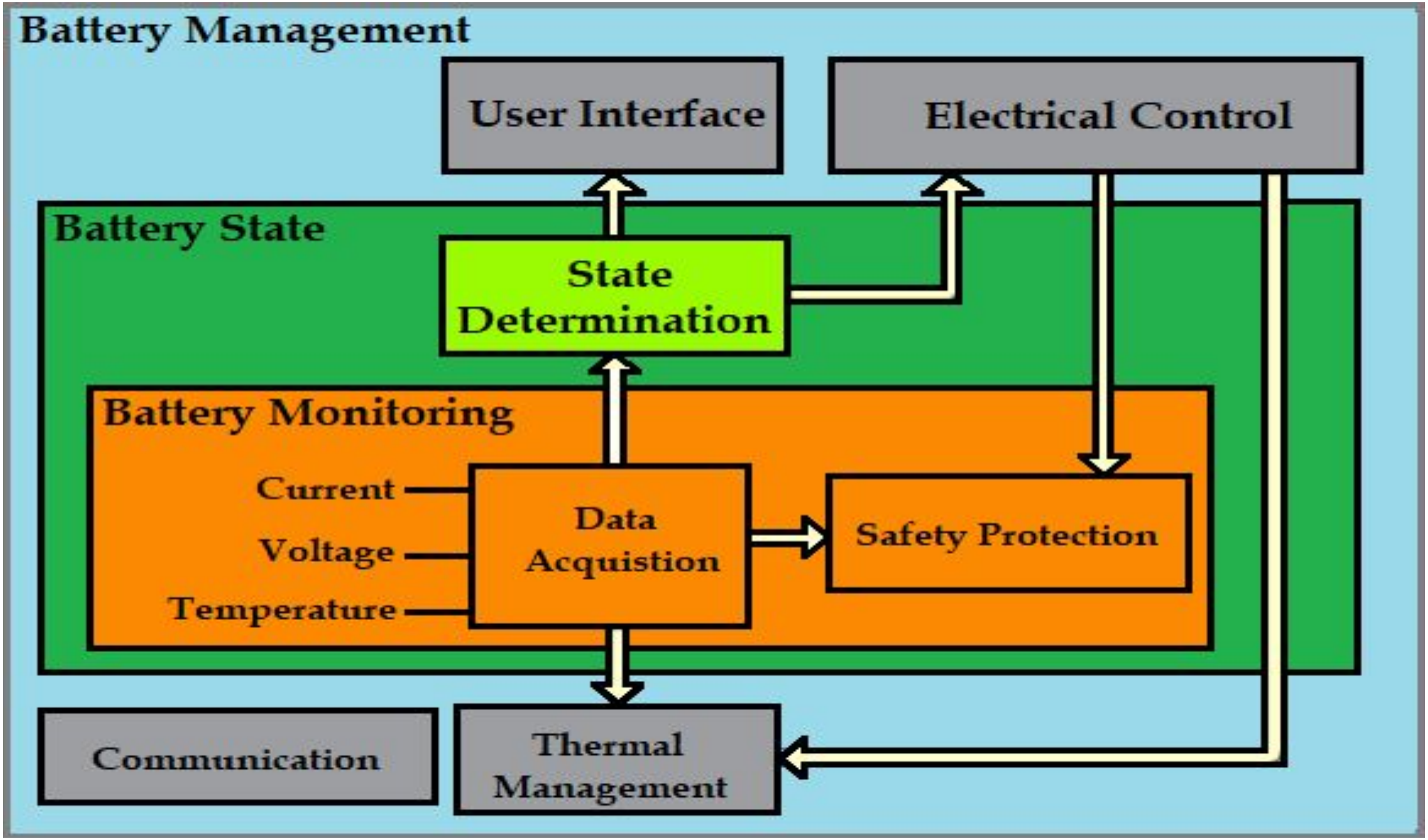
Fuel Gauging = technology to report battery operational status and predict battery capacity under all system active and inactive conditions.

Key benefits are providing extended RUN TIME and LIFE TIME!

The Gas Gauge function autonomously reports and calculates:

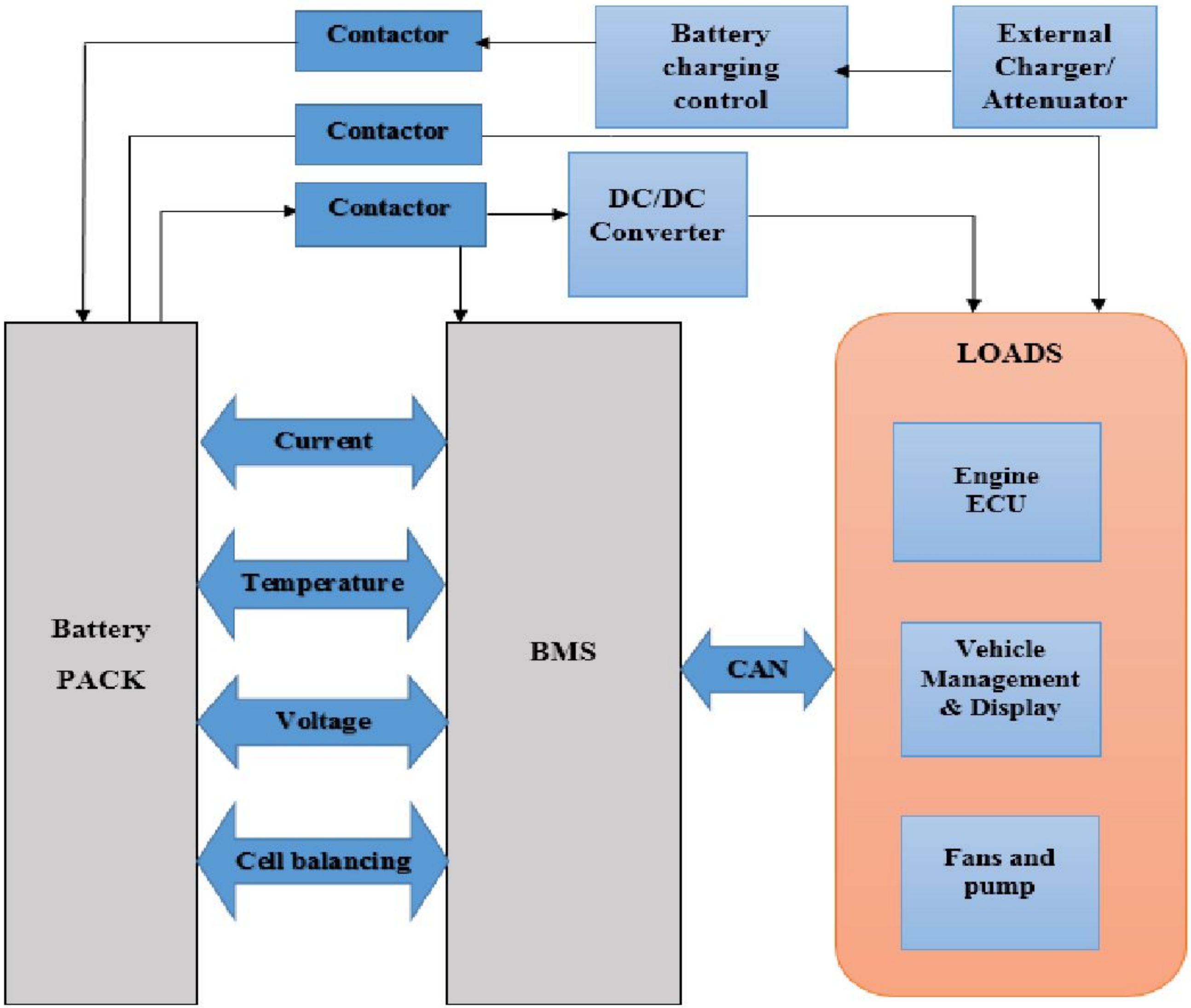
- Voltage
- Charging or Discharging Current
- Temperature
- Remaining **battery capacity** information
  - *Capacity percentage*
  - *Run time to empty/full*
  - *Talk time, idle time, etc.*
- Battery **State of Health**
- Battery **safety** diagnostics

Battery Management



BMS

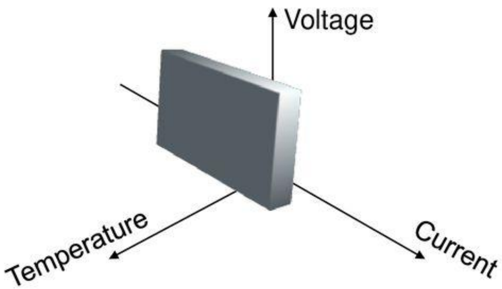
Battery Management



BMS: Process

BMS MAIN FUNCTION: PROTECTION

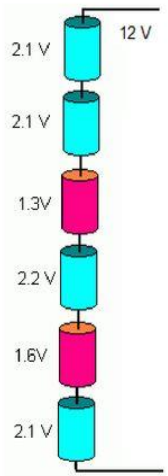
A BMS keeps **EACH** cell within its SOA



BATTERY PROTECTION

Protecting a single cell is hard enough

Protecting a battery
(a series string) is harder:
cell voltages do not divide equally,
temperatures vary



BMS 2nd FUNCTION: BALANCING

- All cells equally charged = maximum available energy
- Balancing removes charge from fullest cells, to leave room for more charging, so the other cells can catch-up

DIGITAL BMS: MUCH MORE

Evaluation of State of Charge (“Fuel Gauge”)

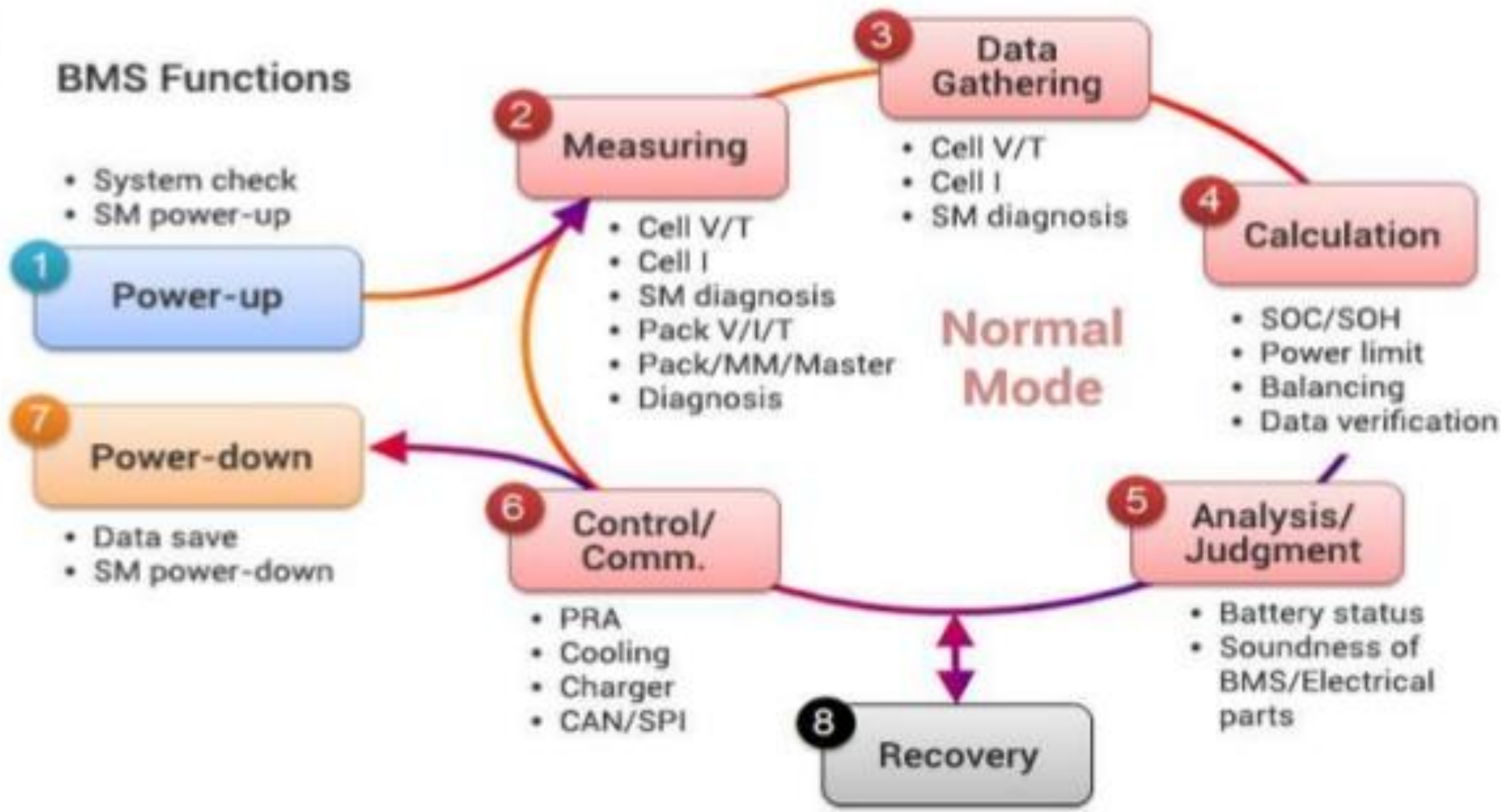
Evaluation of State Of Health

Knows what, where and by how much

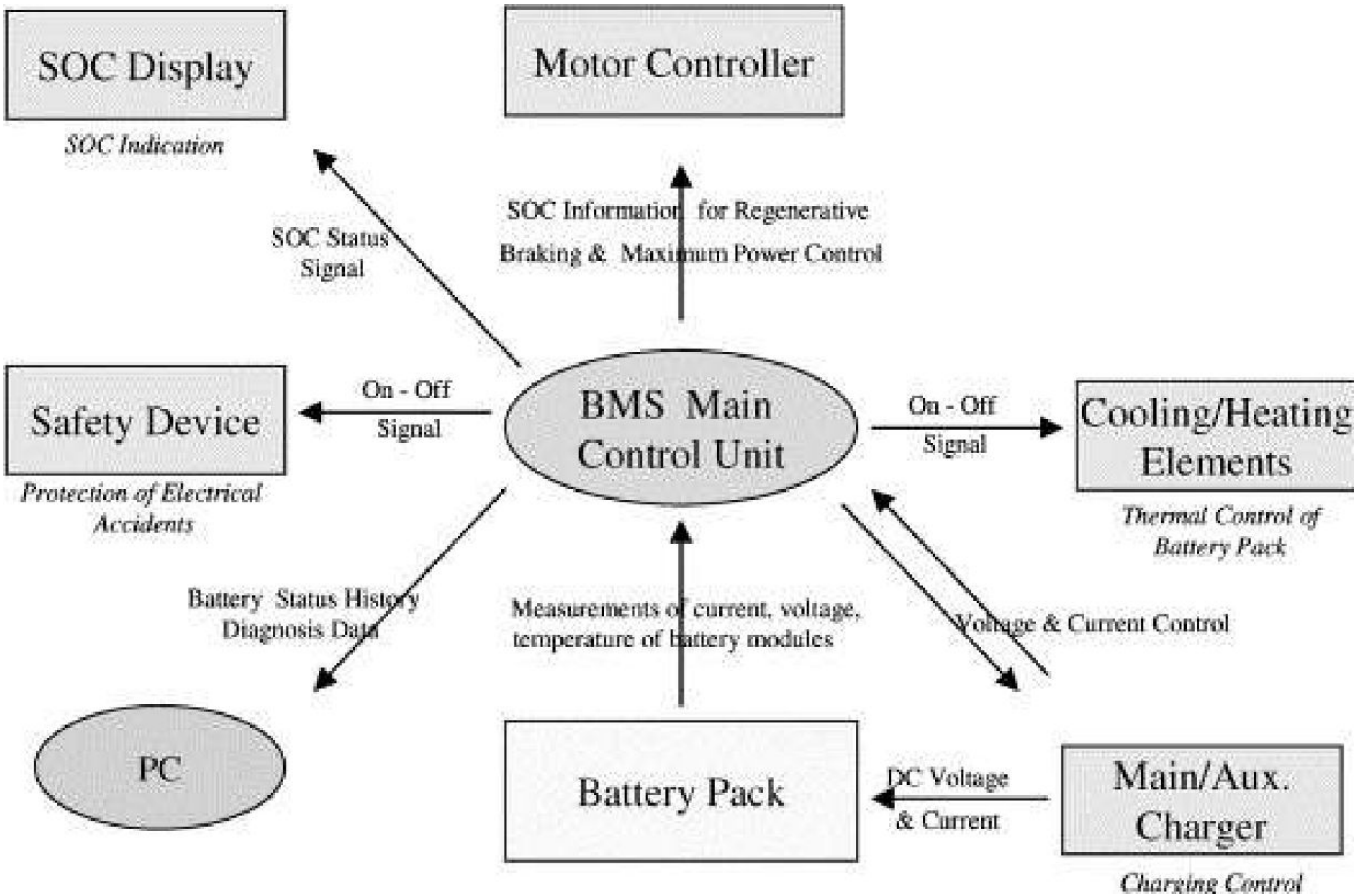
Reports

Requests shut down (doesn't include switch)

BMS Functionality



BMS Key Functioning



Battery Maintenance

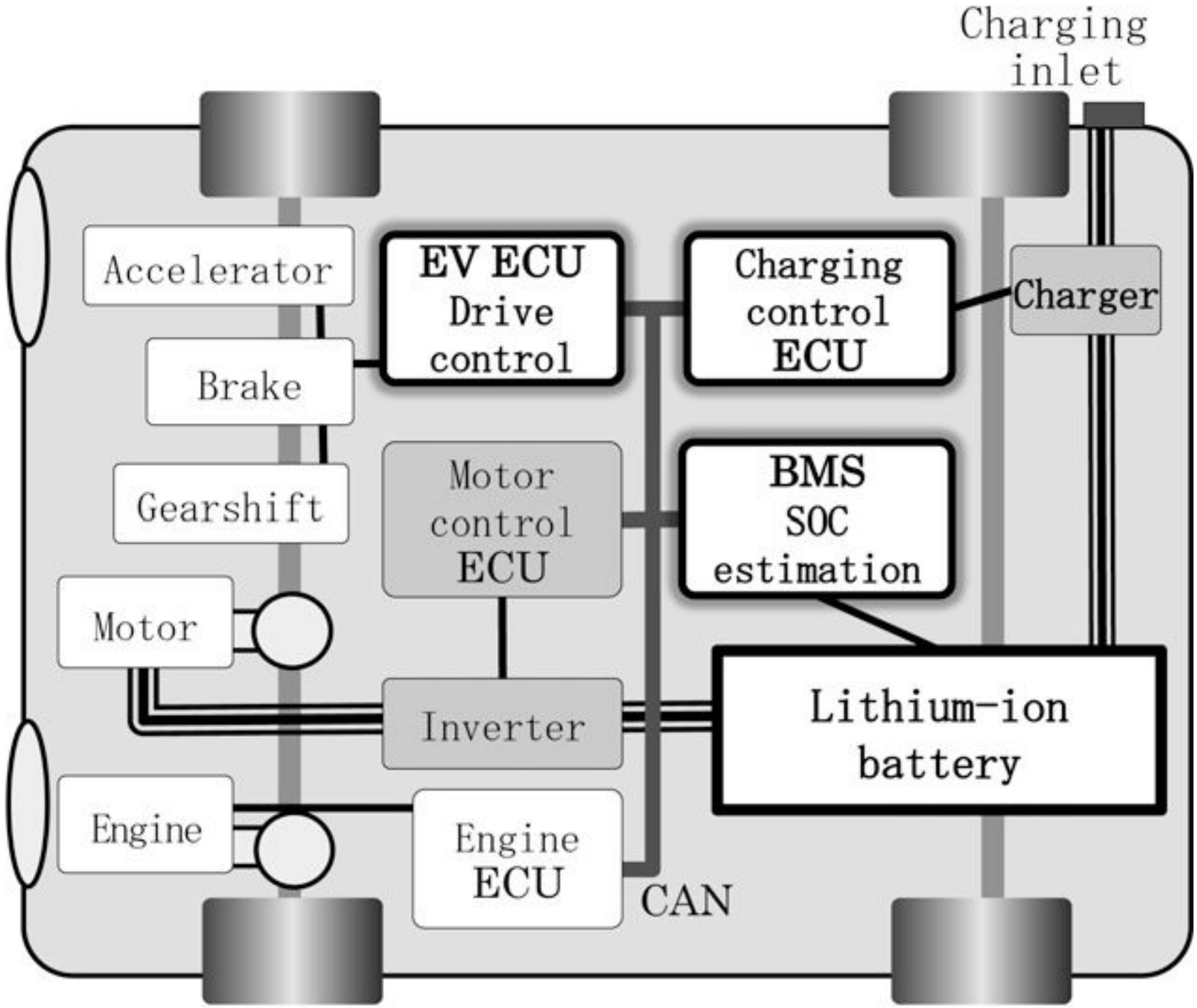


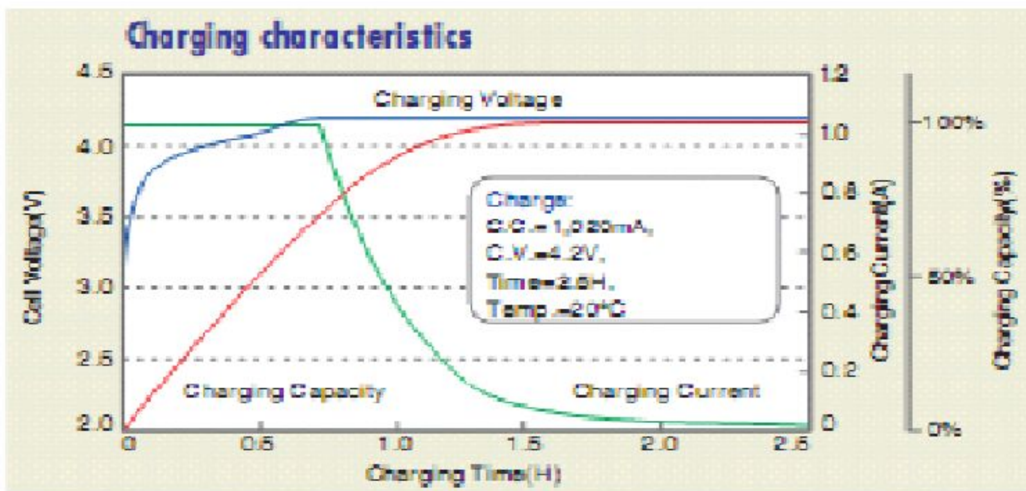
Fig. 1 Block Diagram of Electric Vehicle

State of Charge (SOC)

The State of charge is the available capacity, it also called "Gas Gauge" or "Fuel Gauge" function

Of the various techniques for estimating SOC, two are:

- The battery voltage translation
- The battery current integration ("Coulomb Counting")



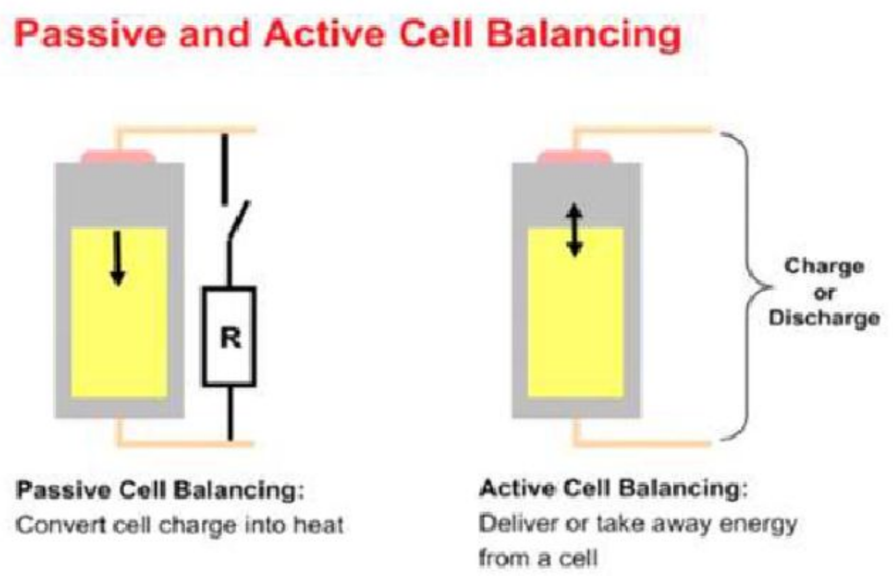
Battery Balancing

Passive balancing

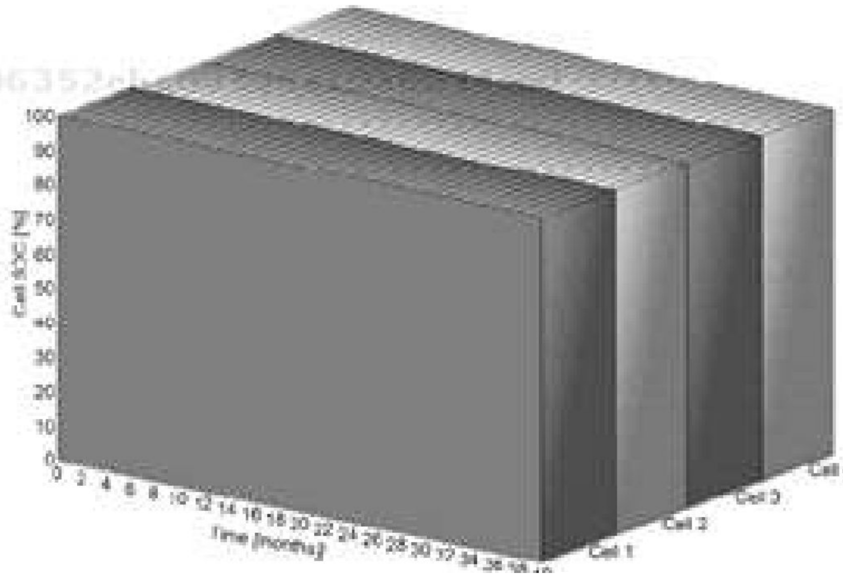
Removing the excedent of charge from a full charged cell using for that purpose a resistor in order to have a match between the cell of the lower cells in the charge reference.

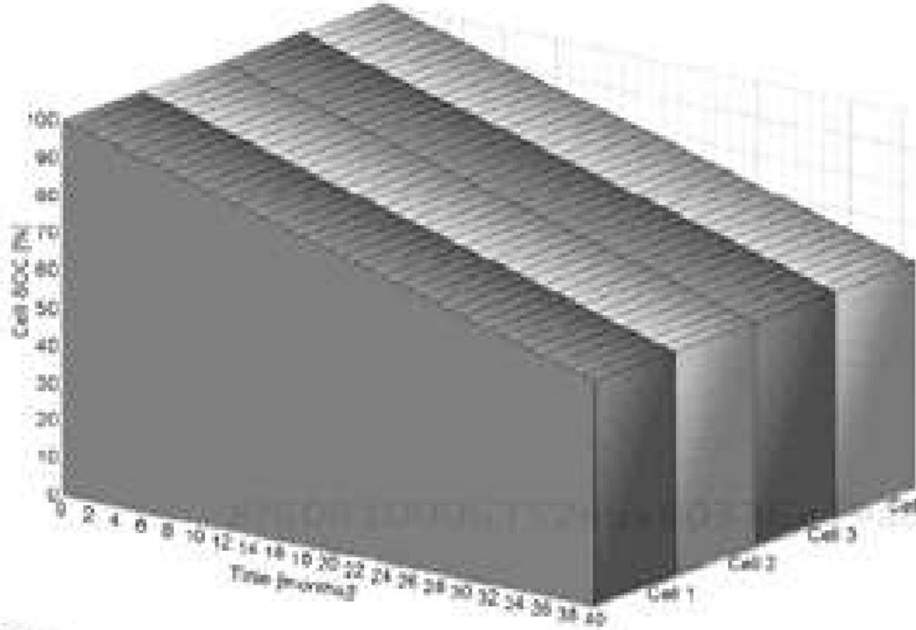
Active balancing

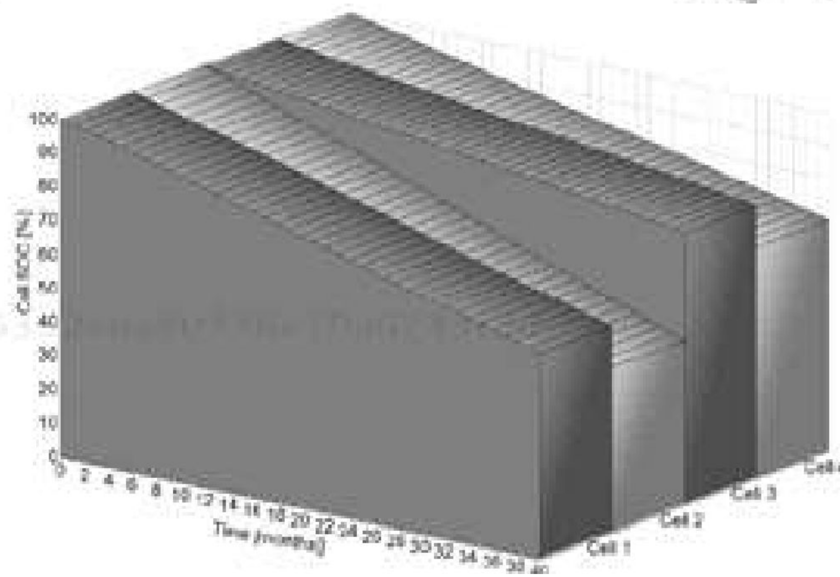
Removing charge from higher energy cells and delivering it to lower energy cells, for that purpose element as capacitor are used for storing the energy.



Cell Balancing





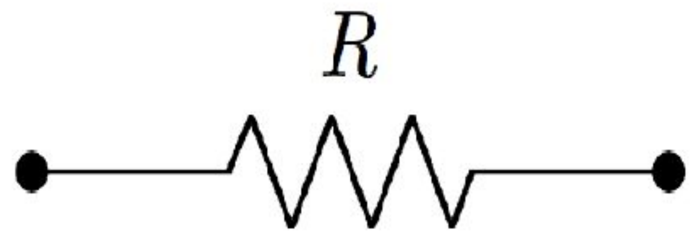


https://www.youtube.com/watch?v=t0YY8t6DKL8

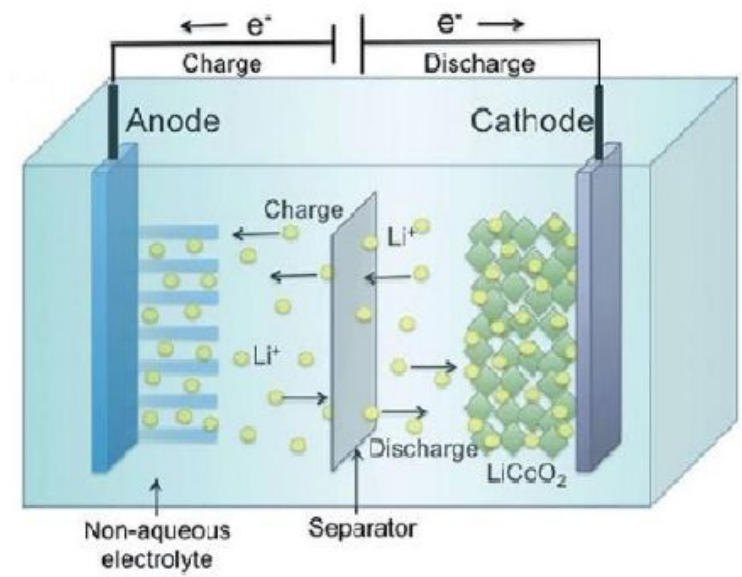
Battery Thermal Management

Heat in a battery cell is produced by:

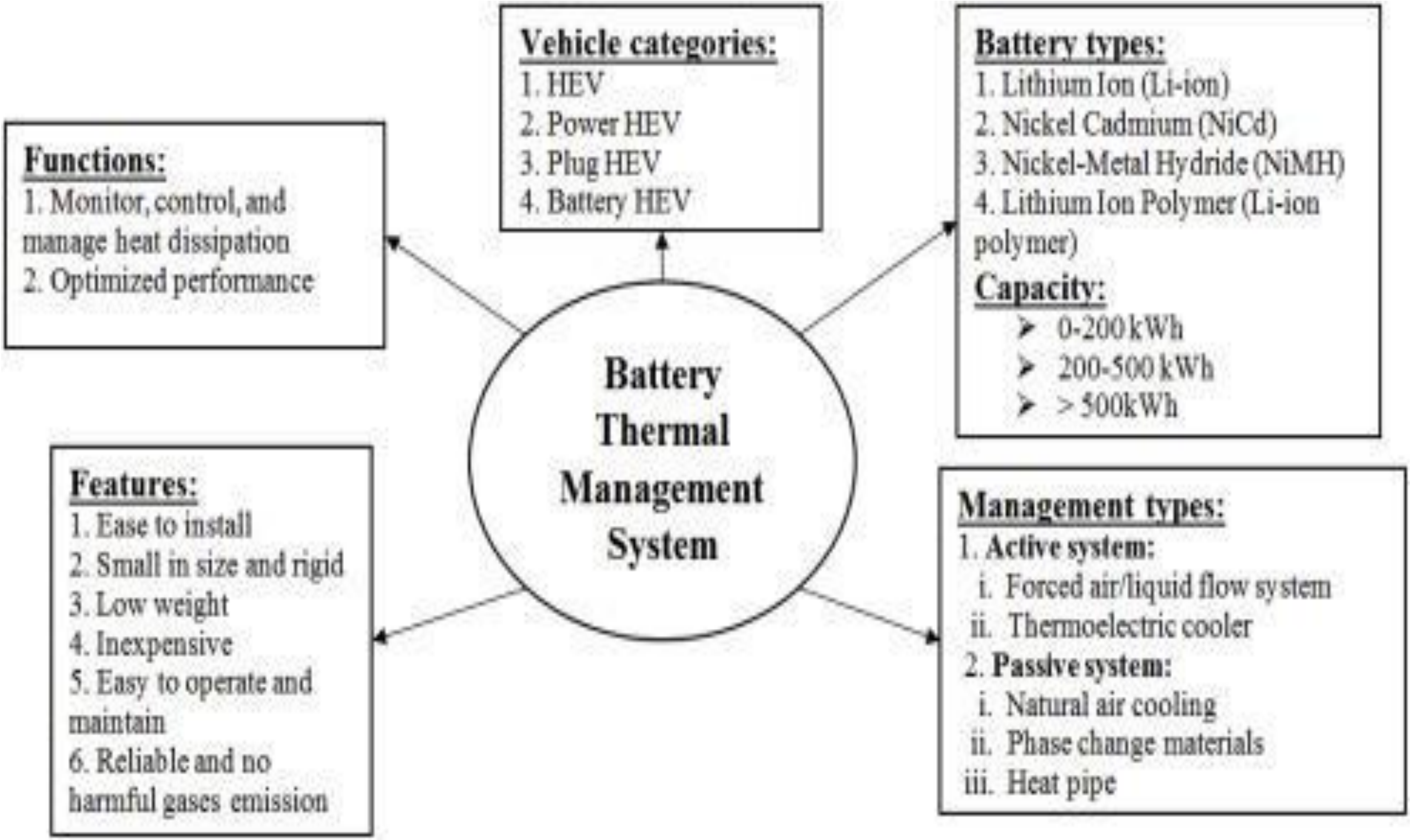
1. The resistance of the various cell components (electrode, cathode, anode, etc.); Joule heating, which can be minimized by cycling the cells at low currents



2. Entropic reactions within the cell (exothermic reactions within the cell due to the transfer of ions and electrons)

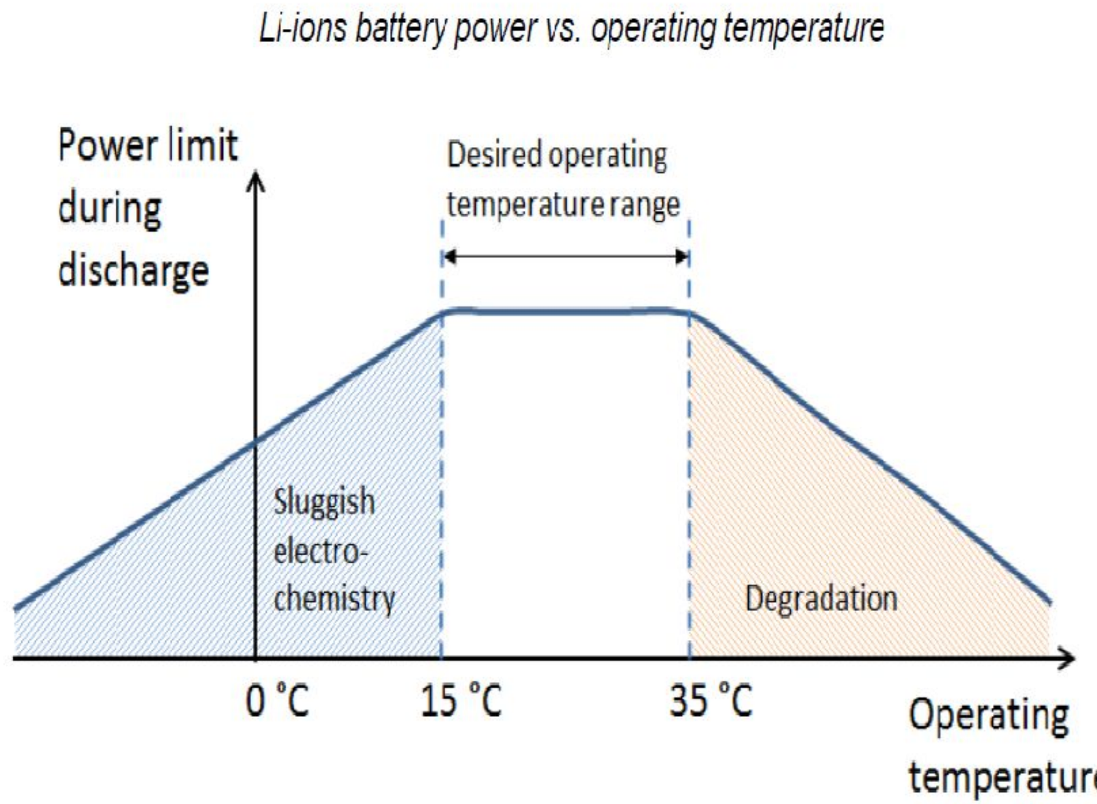


Battery Thermal Management

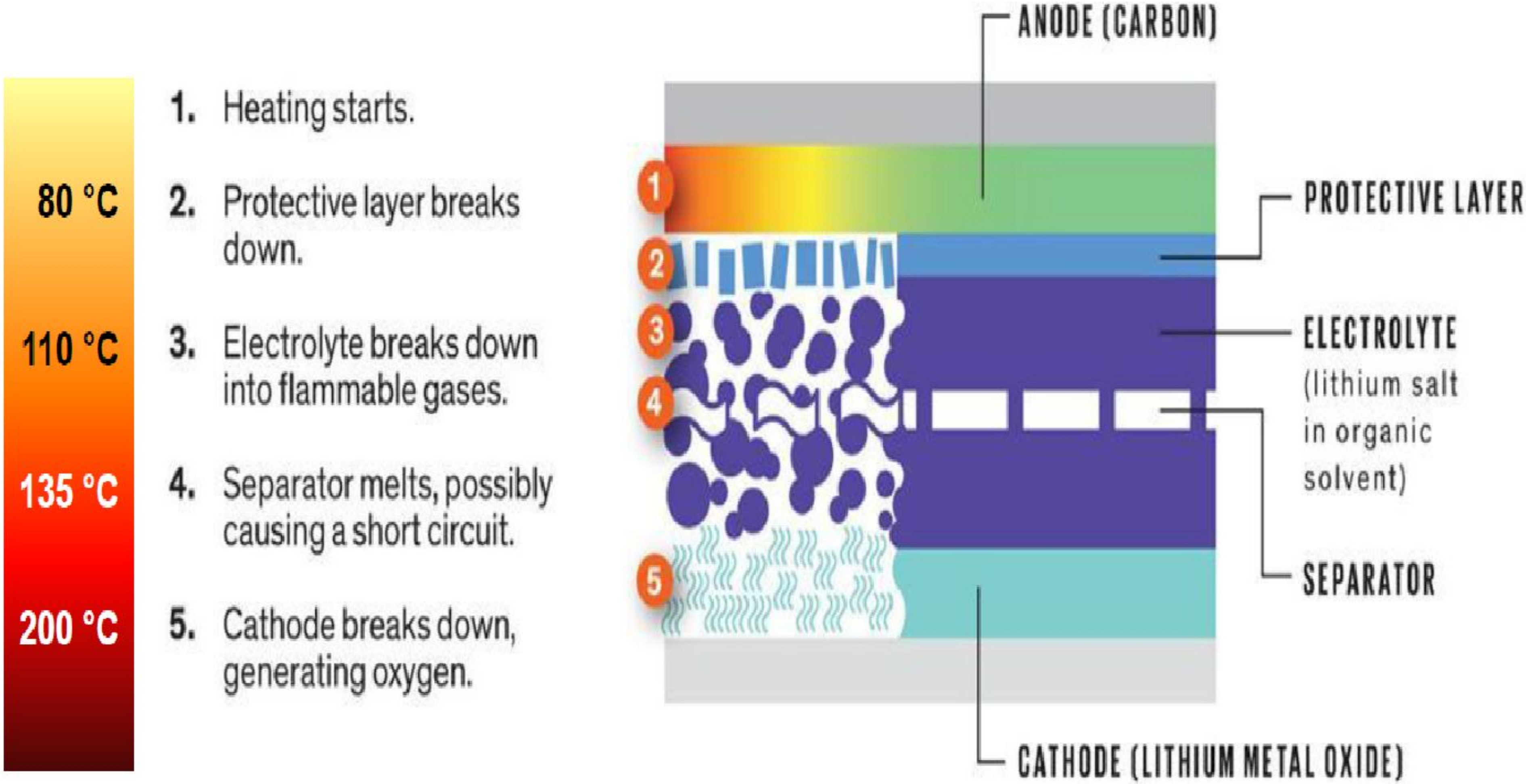


Battery Thermal Management

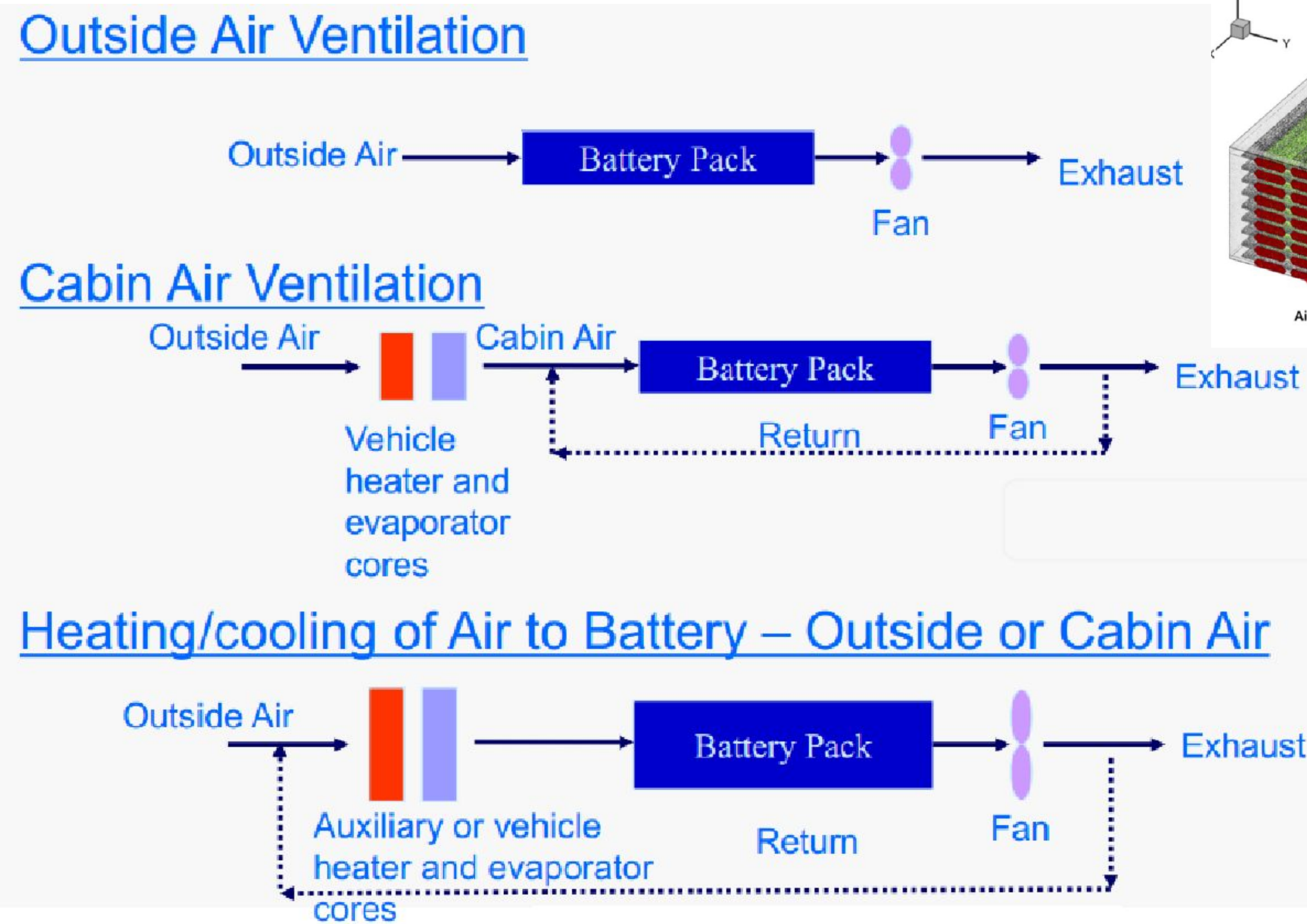
Temperature is one of the most critical factors in designing and operating batteries for HEVs

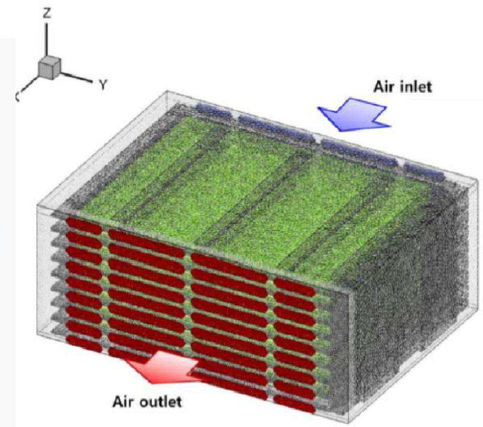


Thermal runaway in lithium battery

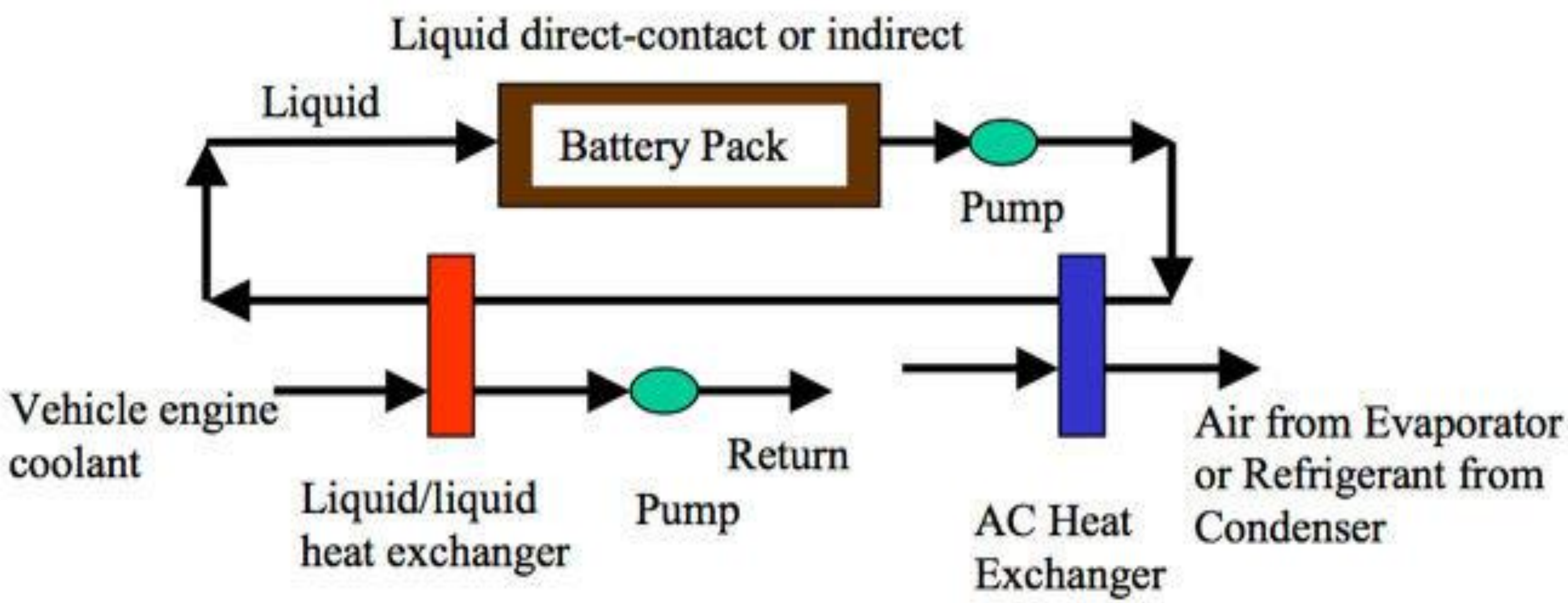


Battery Thermal management-Air cooling

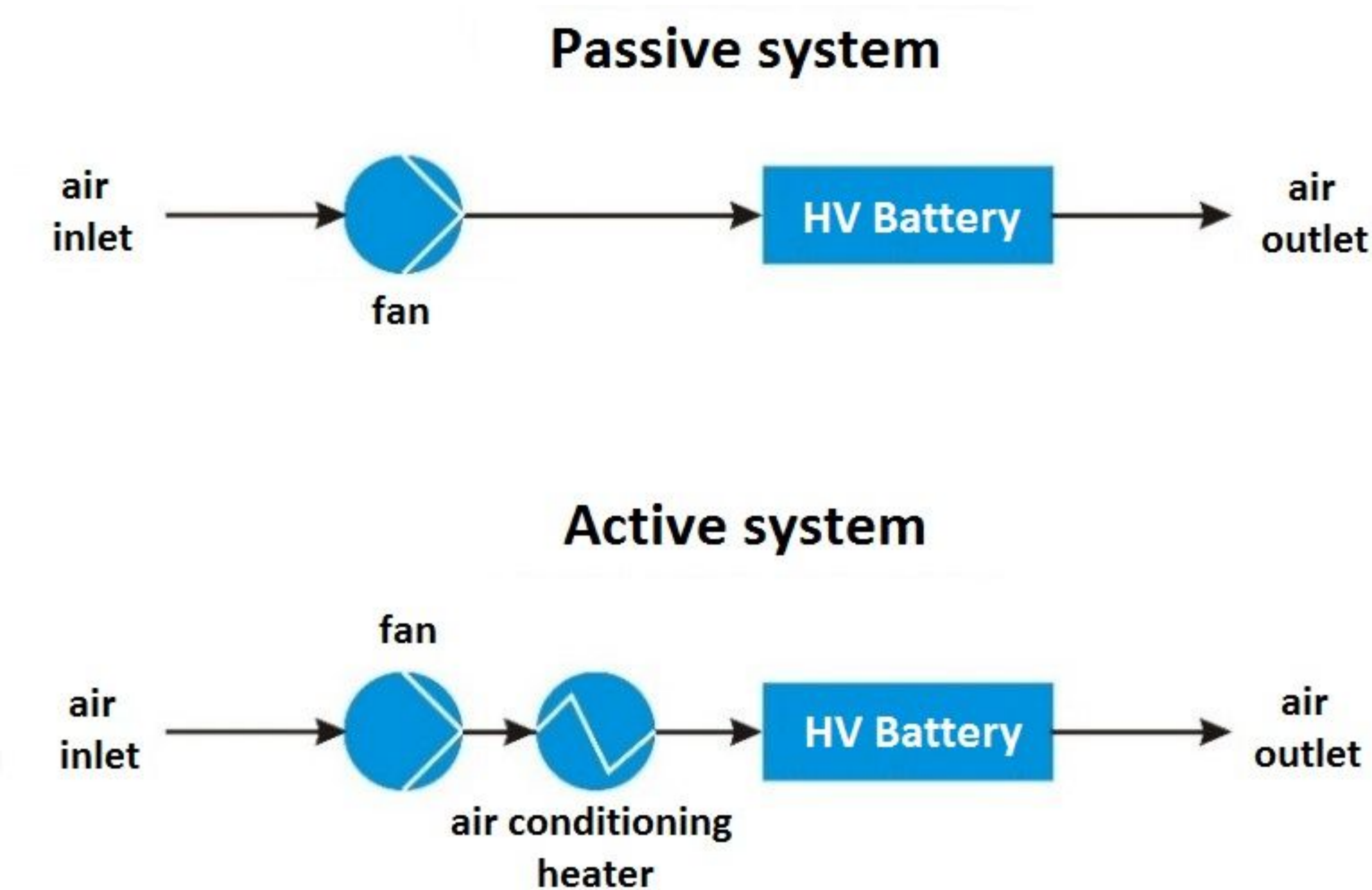




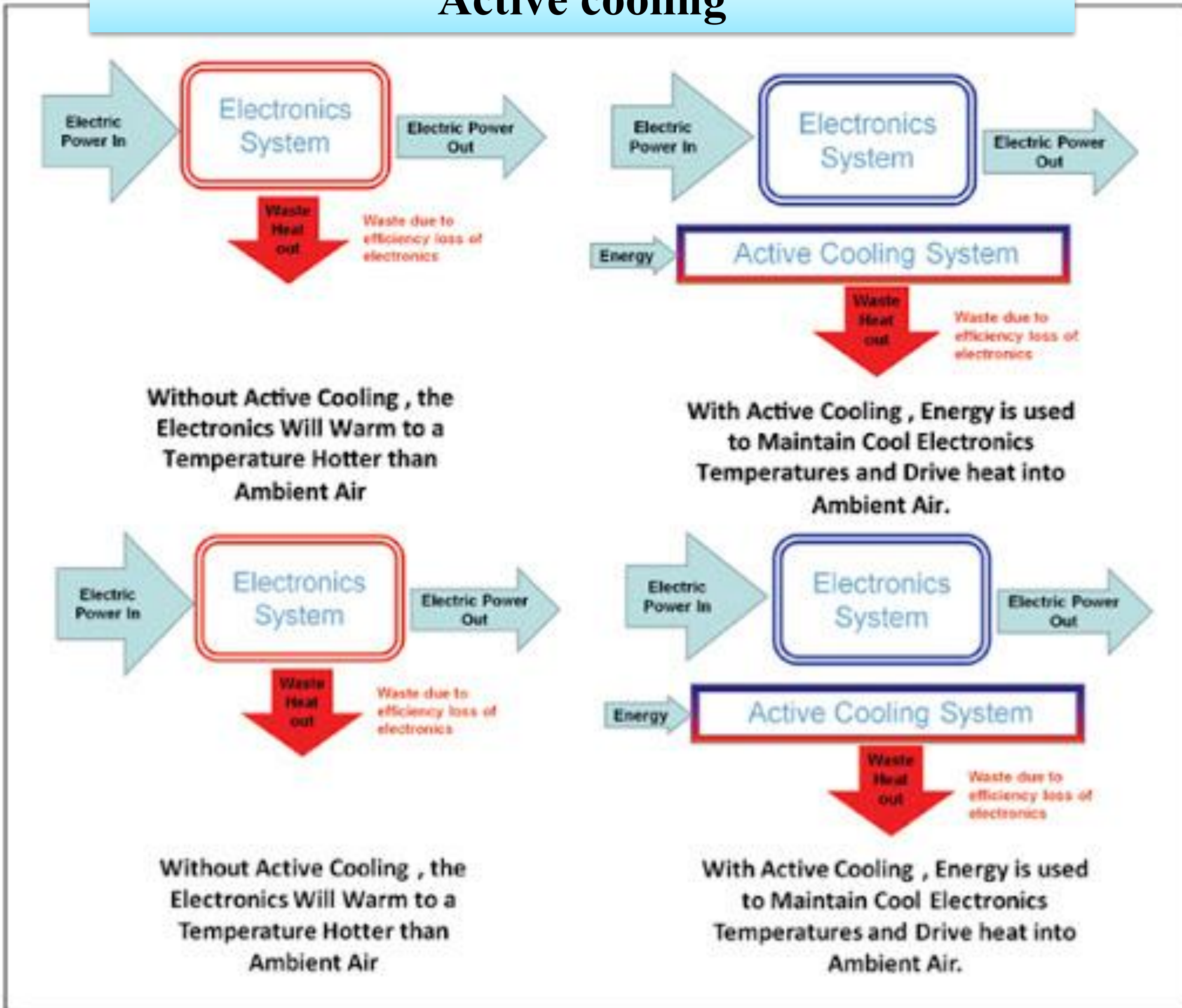
Active cooling



Active cooling



Active cooling

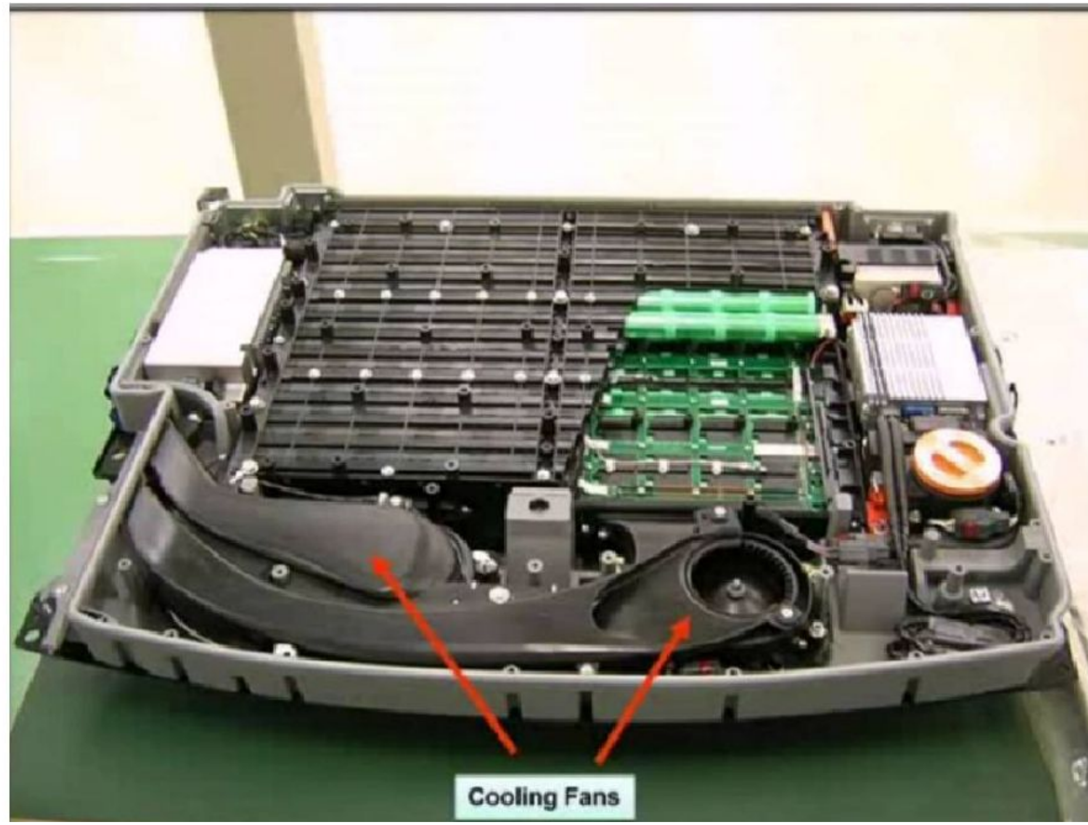


Active air cooling in Ford Escape hybrid 2012

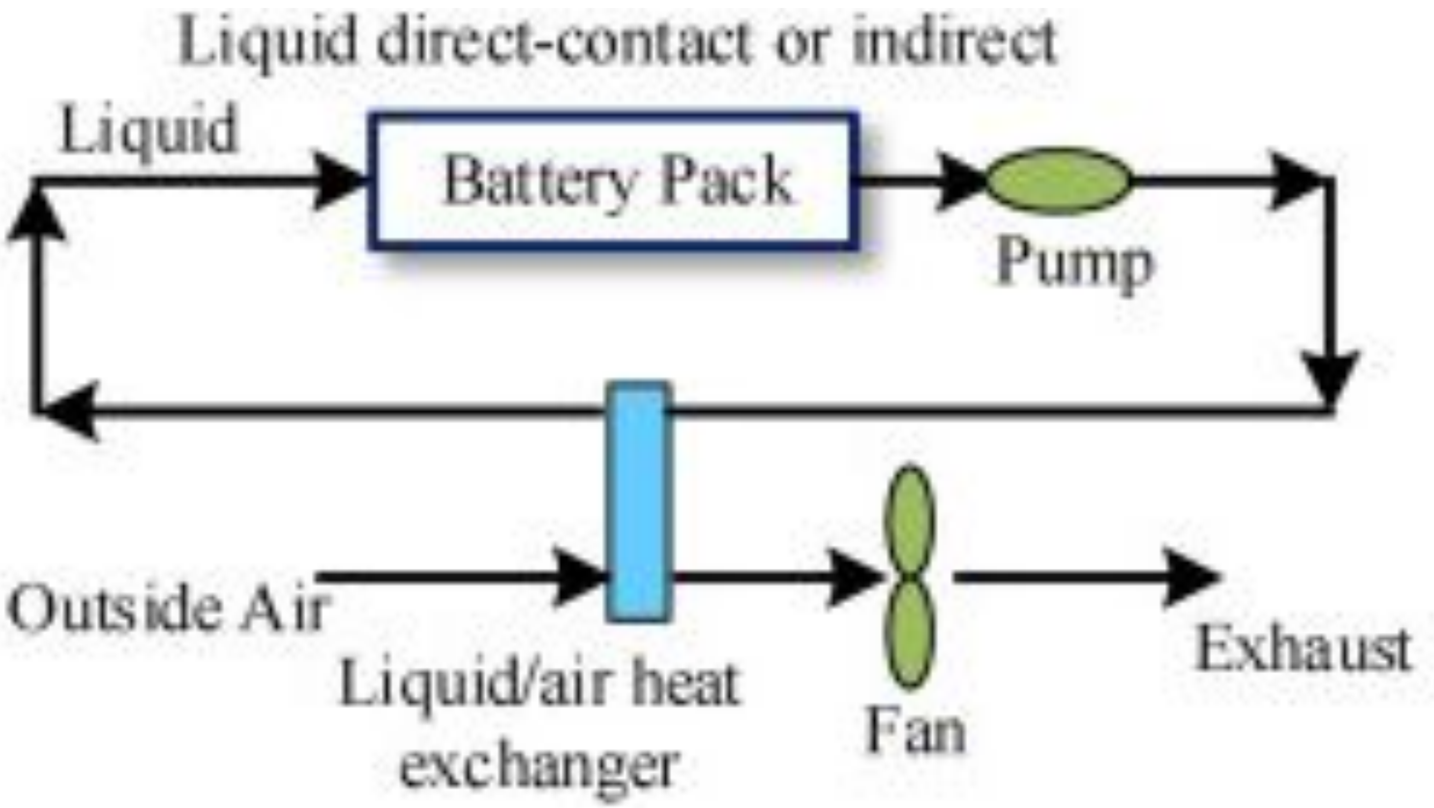
FORD Escape (Plug-in Hybrid)

- 10 kWh Nickel-Metal-Hydride (NiMH) battery pack

- Air cooled



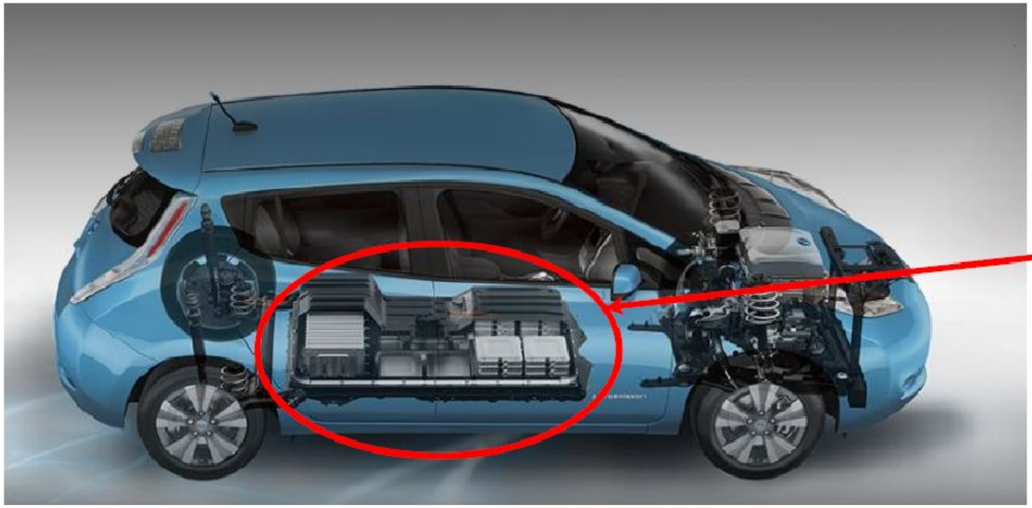
Passive cooling in Nissan Leaf

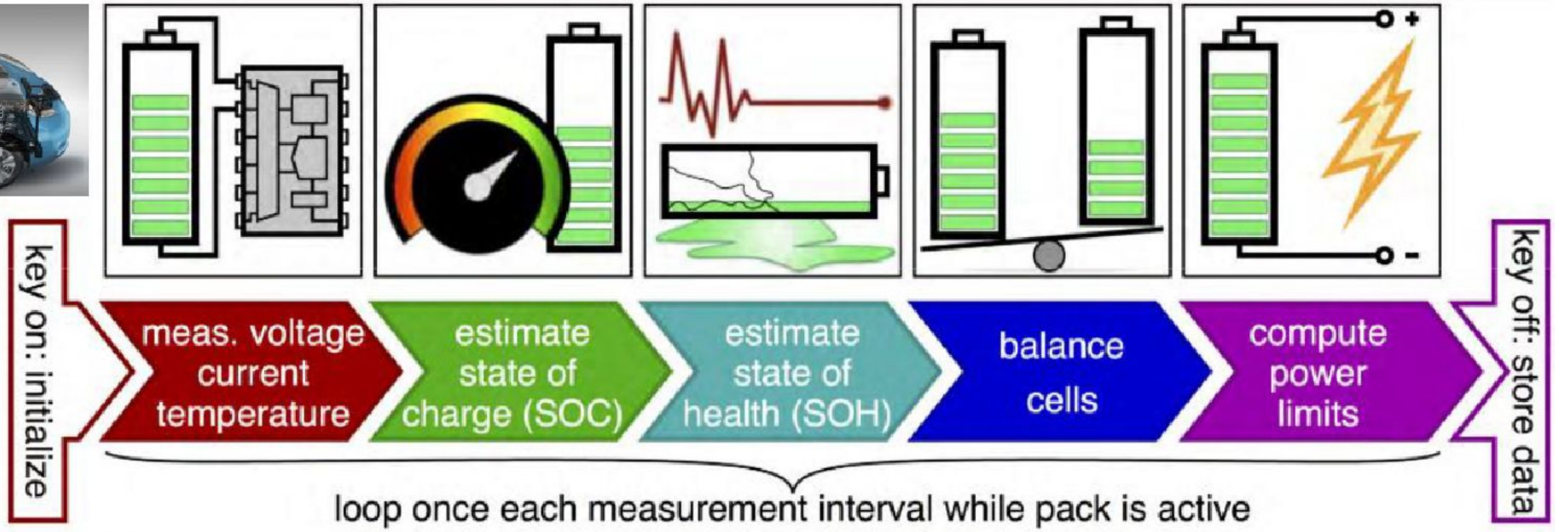


Passive cooling in Nissan Leaf

NISSAN Leaf (Full electric)

- 40 kWh Lithium-ion battery pack.
- 192 Lithium ion cells (48 modules)
- 303 kg total weight

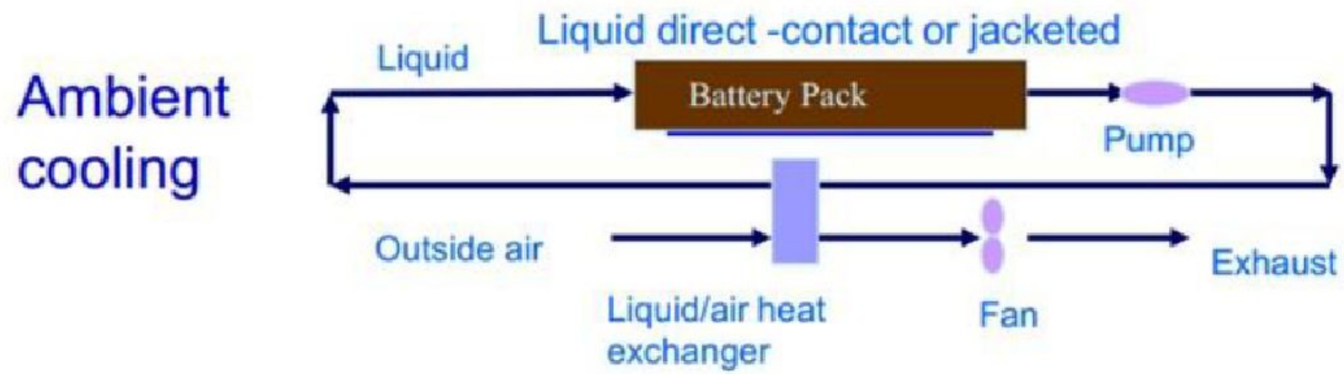


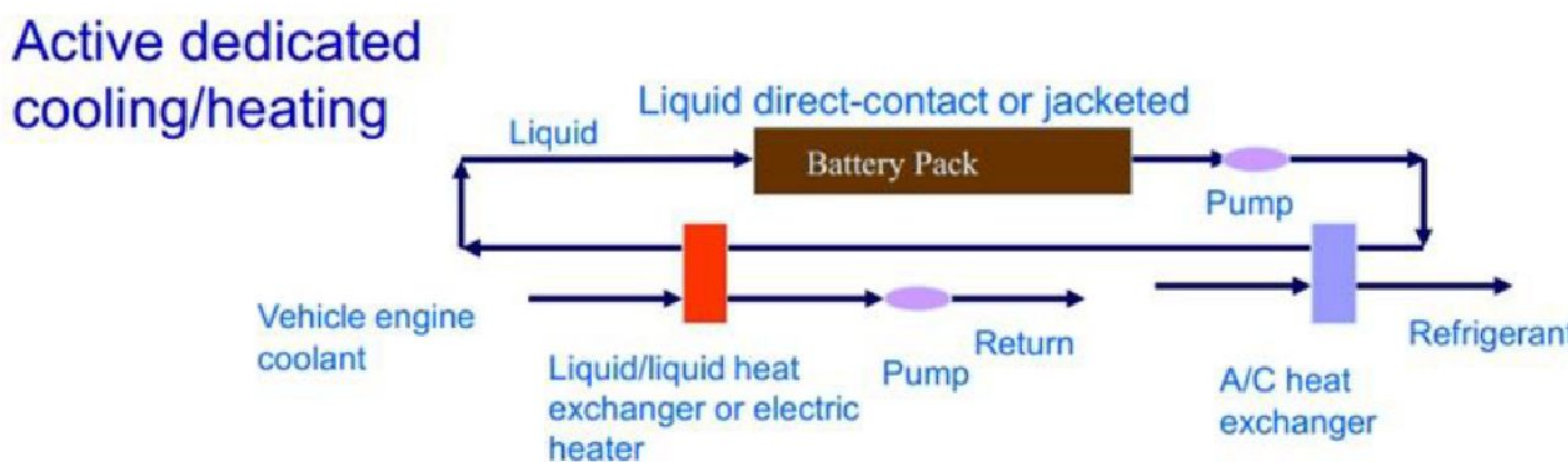


Advantages and Disadvantages of air cooling system

Pro	Con
All waste heat eventually has to go to air	Low heat transport capacity
Separate cooling loop not required	More temperature variation in pack
Low mass of air and distribution system	Connected to cabin temperature control
No leakage concern	Potential of venting battery gas into cabin
No electrical short due to fluid concern	High blower power
Simple design	Blower noise
Lower cost	
Easier maintenance	

Liquid Cooling systems





https://www.youtube.com/watch?v=a1K3yT00ELo

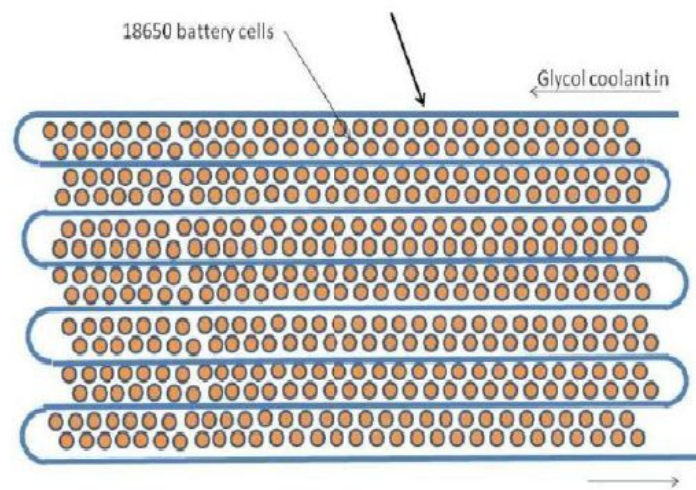
Liquid Cooling systems

TESLA Model S

- 85 kWh Lithium-ion battery pack.
- 7104 CYLINDRICAL Lithium ion cells (16 modules wired in series)
- 504 kg total weight

- Coolant: Water + Ethylene Glycol Mixture

Tesla patent cooling tube

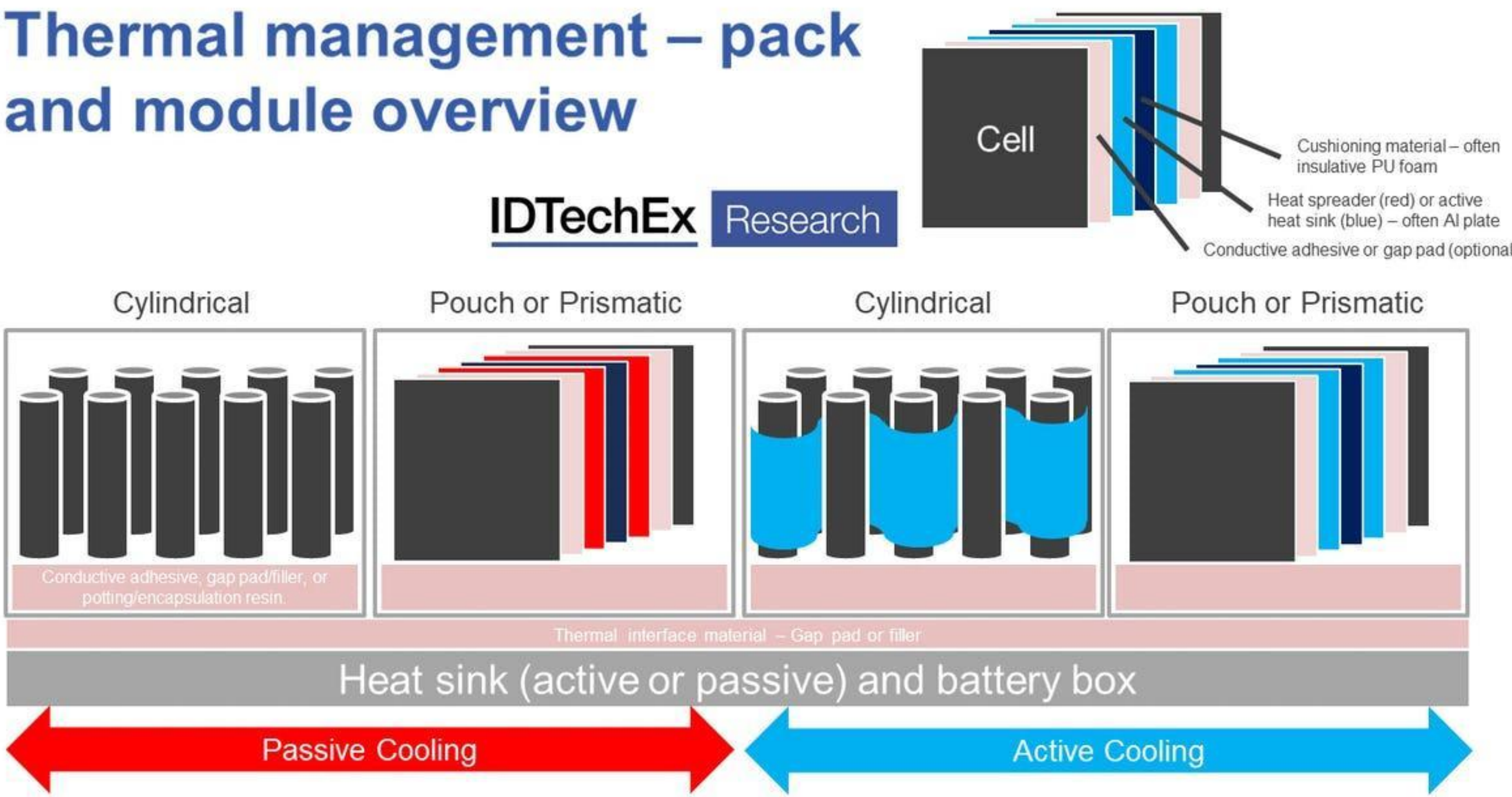


1 Model S battery module (16 total)

Liquid Cooling systems Advantages and disadvantages

Pro	Con
Pack temperature is more uniform - thermally stable	Additional components
Good heat transport capacity	Weight
Better thermal control	Liquid conductivity – electrical isolation
Lower pumping power	Leakage potential
Lower volume, compact design	Higher maintenance
	Higher viscosity at cold temperatures
	Higher cost

Comparison of active and passive cooling system



Battery heating and cooling explained

Informative video on battery thermal management

https://www.youtube.com/watch?v=a1K3yT0OELo

Battery management in Hindi
https://www.youtube.com/watch?v=t0YY8t6DKL8

Tesla battery management
https://www.youtube.com/watch?v=bNd-yJtRPh
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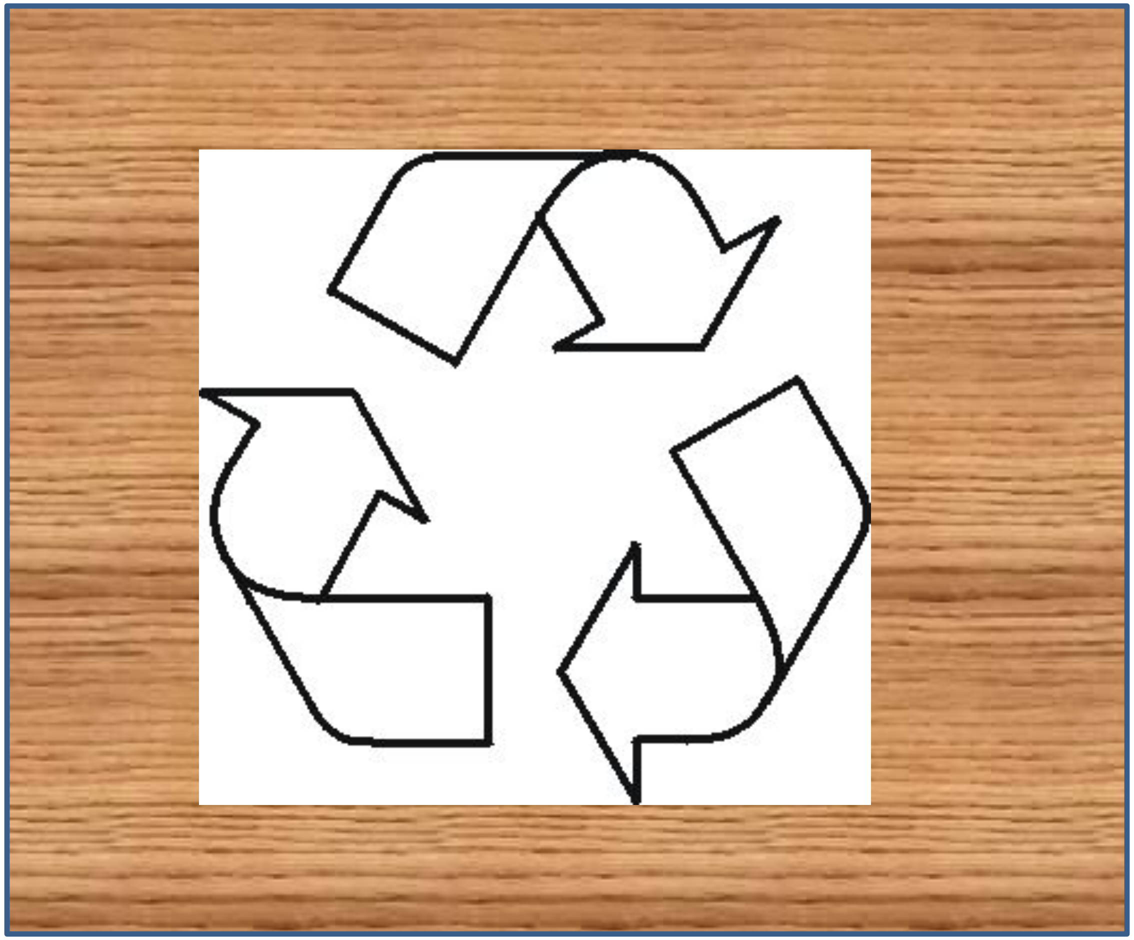
Battery Recycling



Battery Recycling

Why should we recycle?

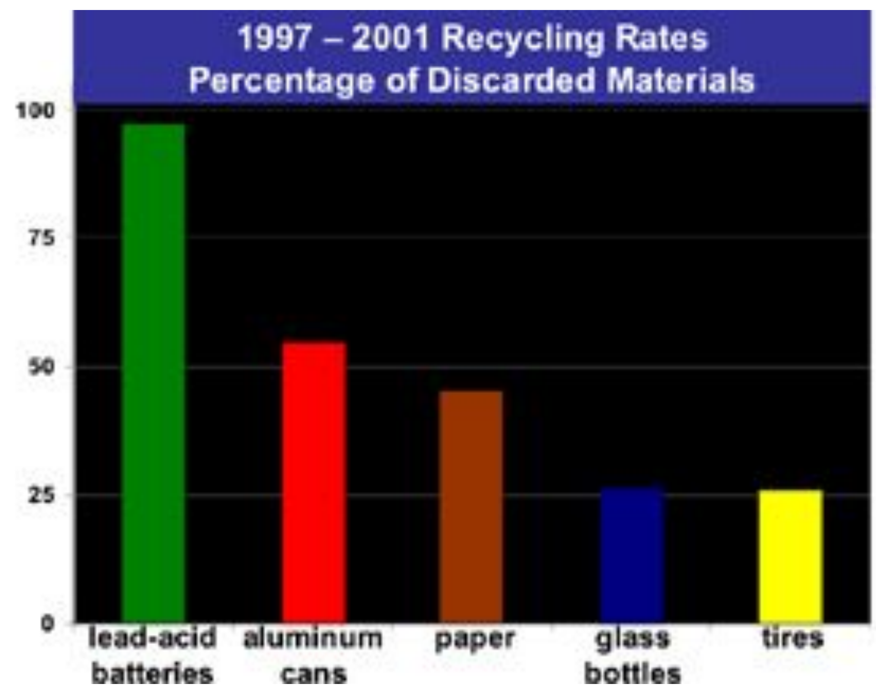
- Where does waste go?
  - Landfills
    - Trash storage
    - Some methane production
  - Incinerators
    - Energy
    - Volume reduction
    - Break down some compounds

Why should we recycle?

- Americans purchase nearly 3 billion dry-cell batteries every year.
- 350 million are rechargeable.
- Only 3-5% of primary dry cells are recycled.
- Nearly 99 million wet-cell lead-acid car batteries are manufactured each year.
- A primary battery will only return 1/50 the energy used to make it.
- Batteries contain heavy metals such as mercury, lead, cadmium, and nickel, which can contaminate the environment when batteries are improperly disposed of.
- The oceans are starting to show elevated levels of cadmium.
- NiCd batteries account for 75% of cadmium in landfills.
- Preserve natural resources.

Battery Recycling

- Companies claim it is not economically profitable.
  - Transportation
  - Sorting
- Lead acid batteries are the most recycled product.



How Recycling works

- Lead Acid Recycling
  - Plastic is broken up and reused
  - Lead and lead oxide is smelted
  - Sulfuric acid is reclaimed

Battery Recycling

Pyrometallurgical and Hydrometallurgical extraction

- Allows metals to be reclaimed from oxides.
- $PbO(s) + CO(g) \rightarrow Pb(s) + CO_2(g)$
- Hydrometallurgical extraction
- $Cu^{2+}(aq) + H_2(g) \rightarrow Cu(s) + 2H^+$

Battery Maintenance and Recycling

Recycling other types of batteries

- NiCd : batteries can be reprocessed through a similar thermal technique
- NiMH: The output of this process is a product with high nickel content which can be used in the manufacture of stainless steel.
- Li-Ion :currently reprocessed through pyrolysis (heat treatment) with the primary recovery the metal content.
- Zinc-carbon/air and alkaline-manganese: can be reprocessed using a number of different methods, which include smelting and other thermal-metallurgical processes to recover the metal content (particularly zinc).
- Batteries containing mercury: Most commonly processed using a vacuum-thermal treatment

Battery Recycling

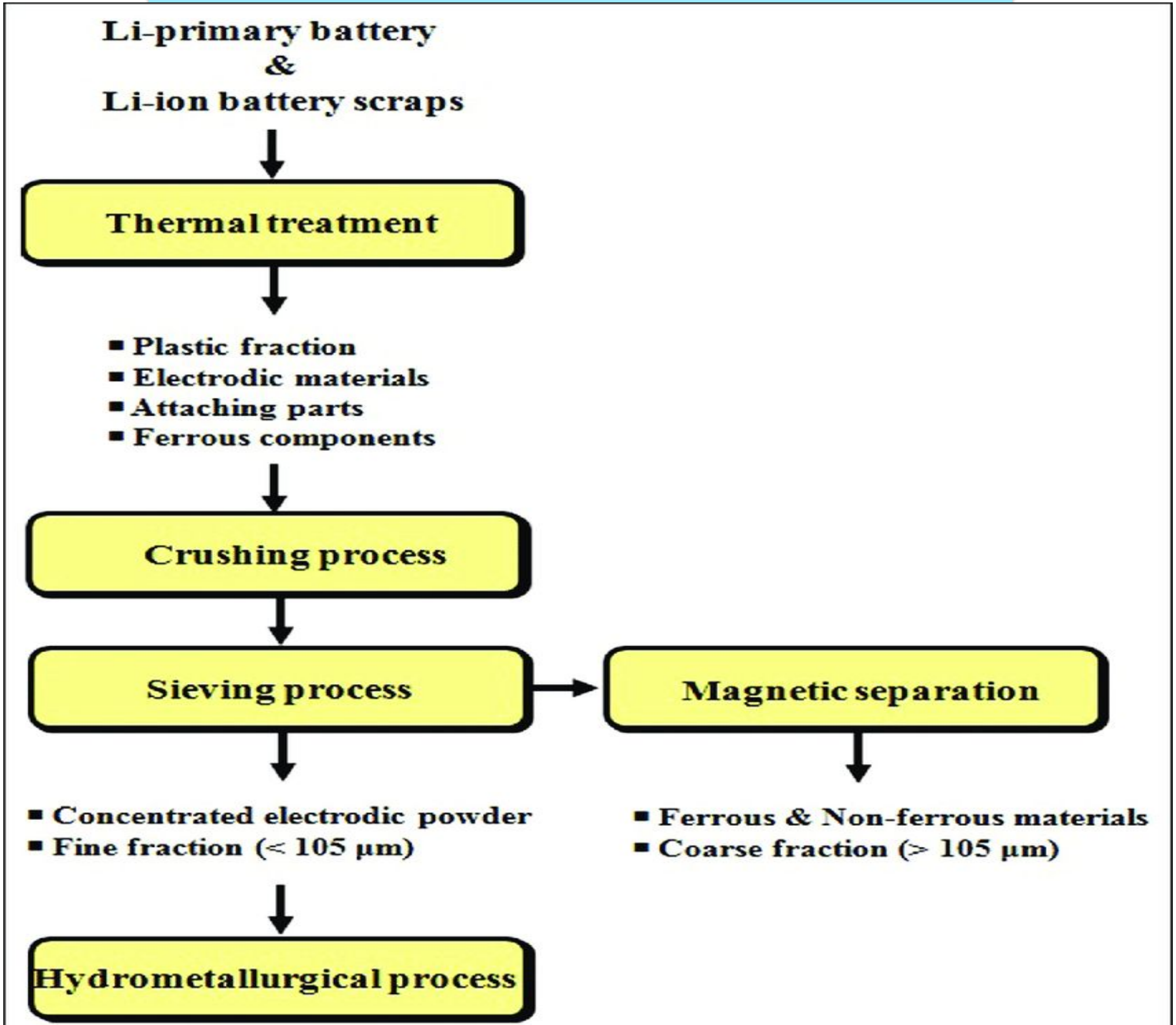
Solutions

- Purchase rechargeable batteries.
- Use solar power
- Battery Deposit
- Make companies take back products they sell.
- Make batteries easier to recycle.

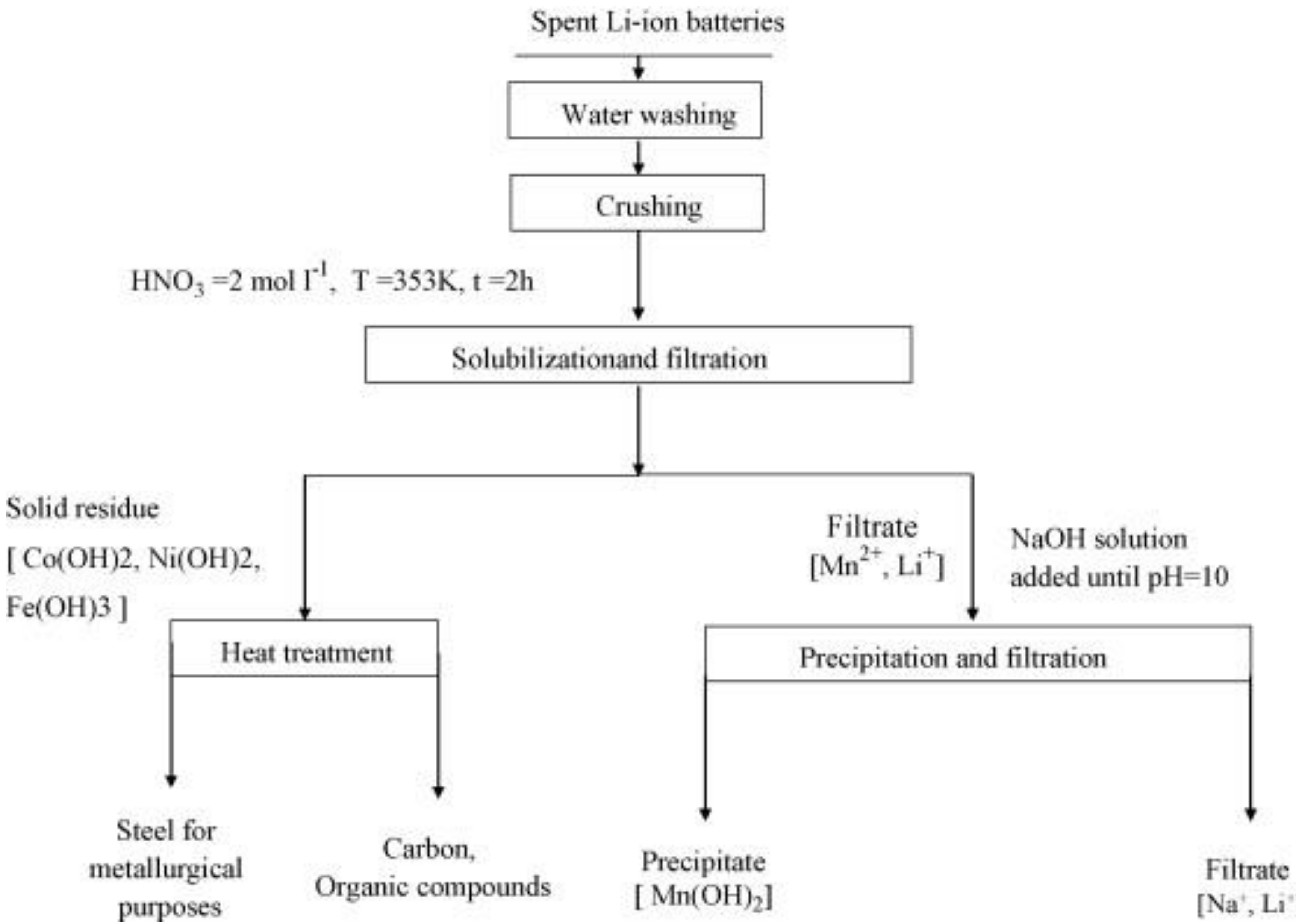
Recycling of mobile phone batteries

Recycling of mobile phone batteries

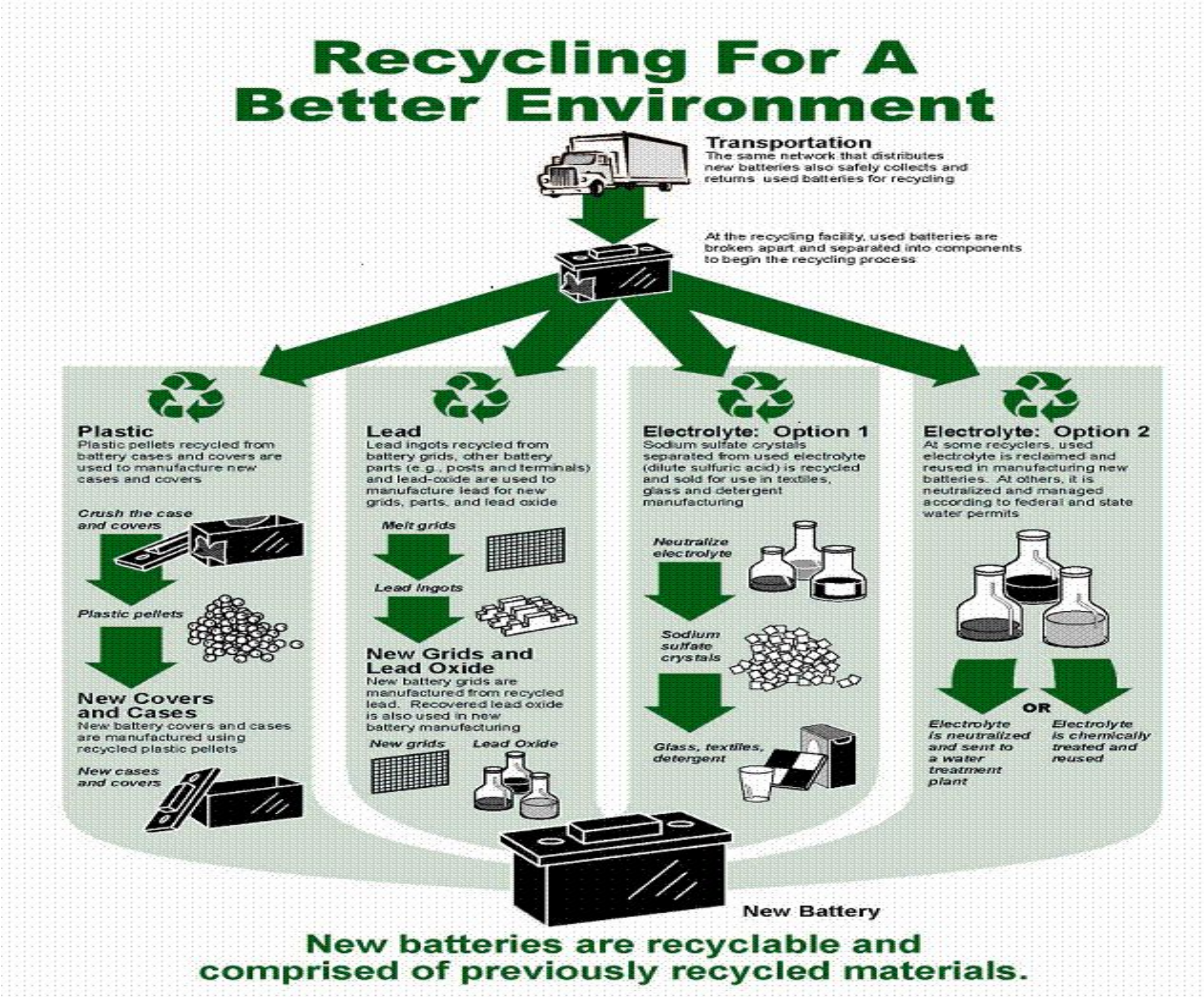
Recycling of lithium ion batteries



Recycling of lithium ion batteries



Battery Recycling



Lead is harmful

Why lead is harmful?

- Lead gets into a person's system, it is distributed throughout the body just like helpful minerals such as iron, calcium, and zinc.
- And lead can cause harm wherever it lands in the body.
- In the bloodstream, for example, it can damage red blood cells and limit their ability to carry oxygen to the organs and tissues that need it, thus causing anemia.
- Most lead ends up in the bone, where it causes even more problems.

Short term lead poisoning

The short-term effects of lead poisoning include:

- Acute fever,
- Convulsions,
- Loss of consciousness and blindness, with anemia,
- Renal failure

Exposure to high levels of lead in a short period of time is called Acute toxicity.

Long term lead poisoning

- Decreased bone and muscle growth
- Poor muscle coordination
- Damage to the nervous system, kidneys, and/or hearing and brain damage.
- Speech and language problems
- Developmental delay
- Seizures and unconsciousness (in cases of extremely high lead levels)

Exposure to small amounts of lead over a long period of time is called Chronic toxicity.

Signs of lead poisoning

- Many kids with lead poisoning don't show any signs of being sick, so it's important to eliminate lead risks at home and to have young kids tested for lead exposure.
- When kids do develop symptoms of lead poisoning, they usually appear as:
  - Irritability or behavioral problems (eating of no nutritious things such As dirt and paint chips)
  - Difficulty concentrating - headaches
  - Loss of appetite - weight loss
  - Sluggishness or fatigue - abdominal pain
  - Vomiting or nausea - constipation
  - Pallor (pale skin) from anemia - metallic taste in mouth
  - Muscle and joint weakness or pain - seizures

Pre cycling steps

Used batteries must be

- *Collected,*
- *Transported*
- *Stored with proper care,*

in order to avoid adverse health effects and environmental contamination.

Acid drainage

Others

Safe Box

The drainage of this liquid may pose several threats to the human health and to the environment, the drainage at collection points it should be avoided;

Rain leaching

Others

Save Box

- the storage facility must be sheltered from rain and other water sources,

Transportation

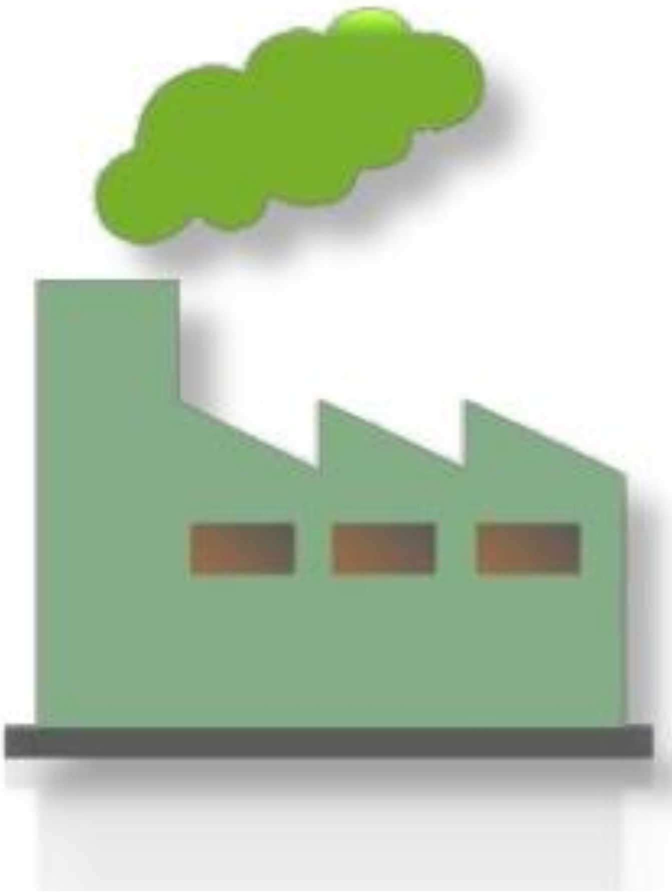
Others

Collection truck

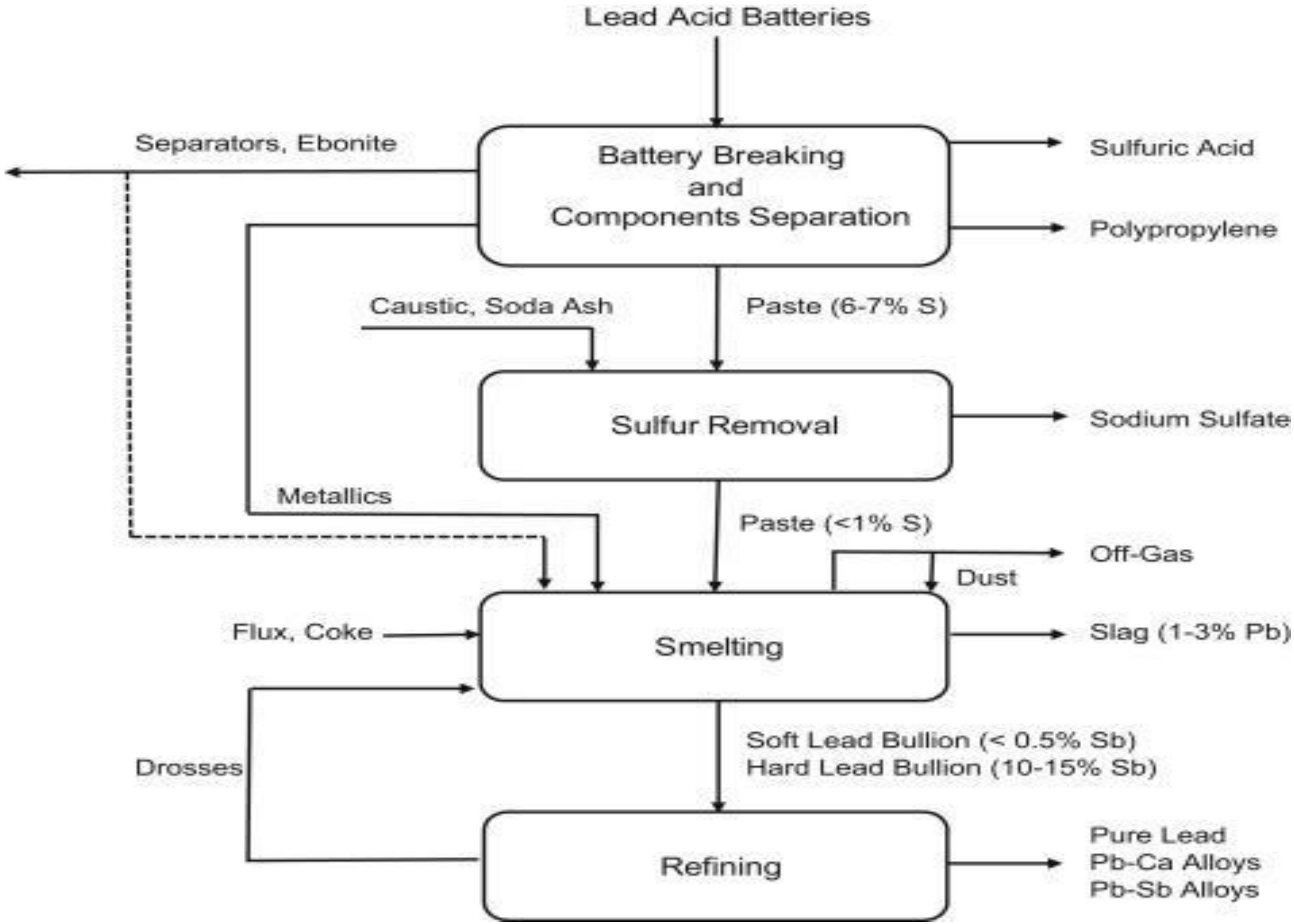
Used lead-acid batteries must be considered as hazardous wastes when transport is needed. drivers and auxiliaries should be trained and use PPE

Plant specification

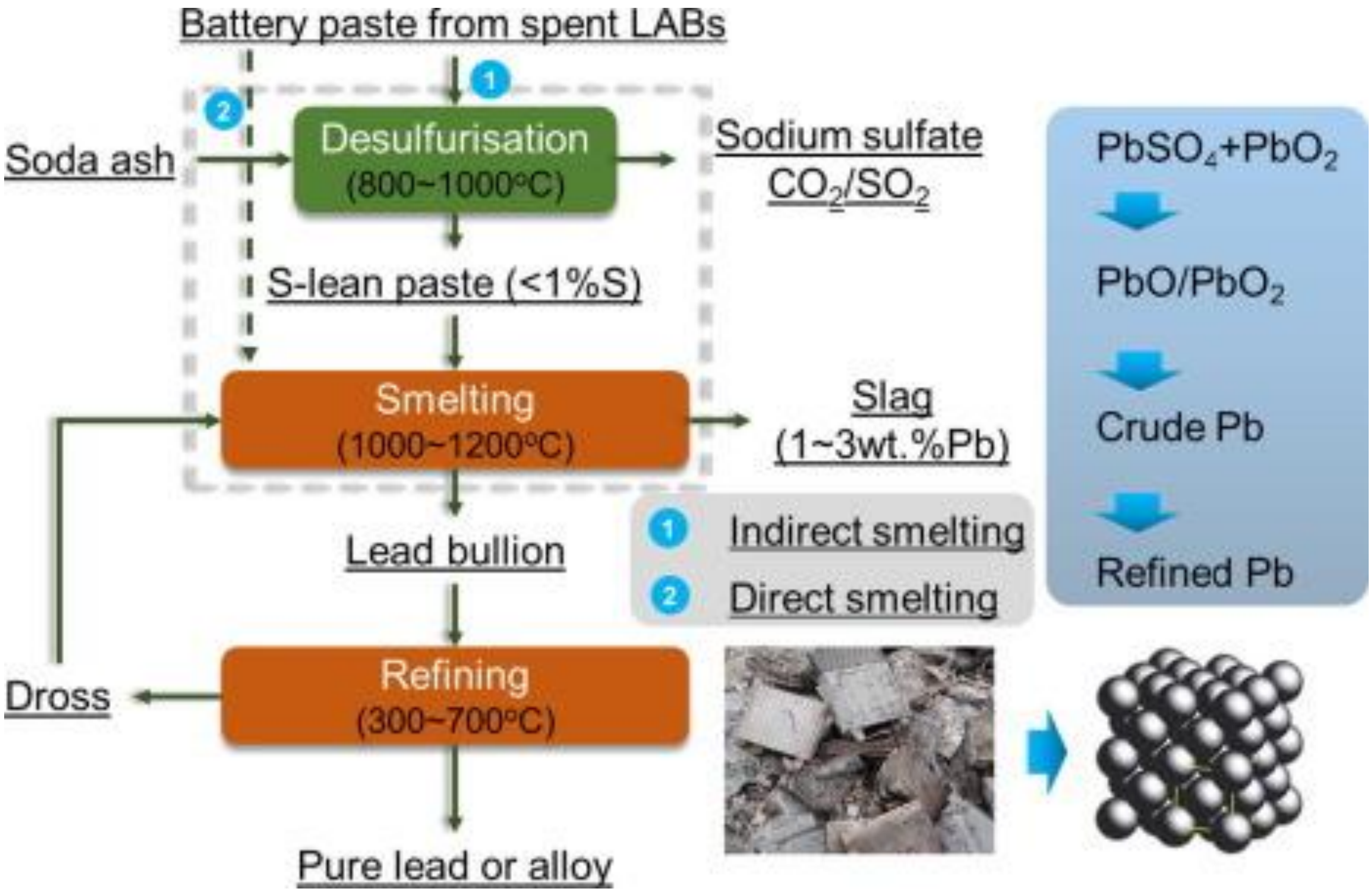
1. Electrolyte Collection and Filtration Unit
2. Breaker and Separation Unit
3. Lead paste Desulphurization &Filtration
4. Sodium Sulphate Production Unit
5. Gaseous Effluents Scrubbing System Unit
6. Lead Smelting Unit
7. Lead Refinery and casting Unit



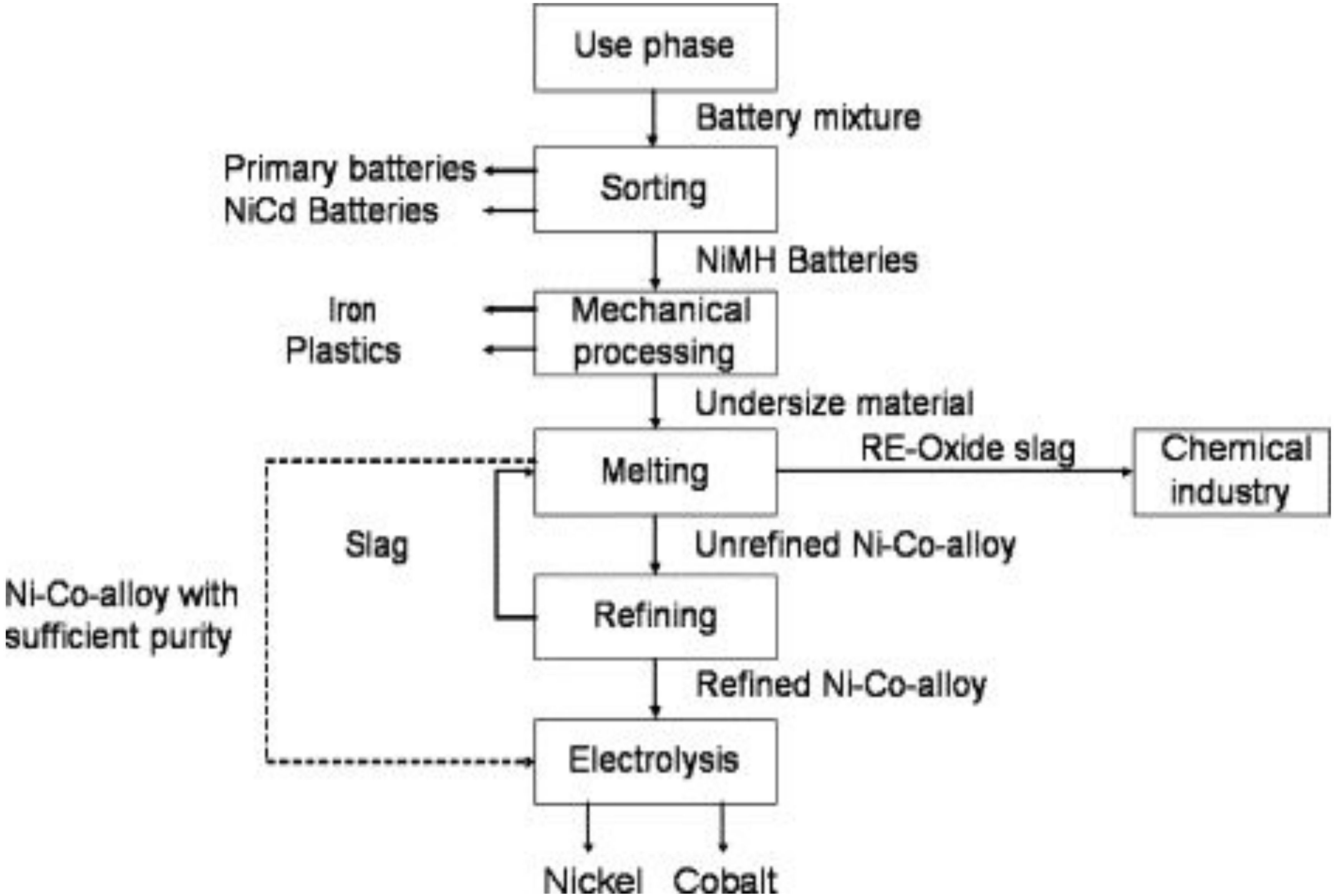
Recycling of lead acid batteries



Lead acid battery Recycling



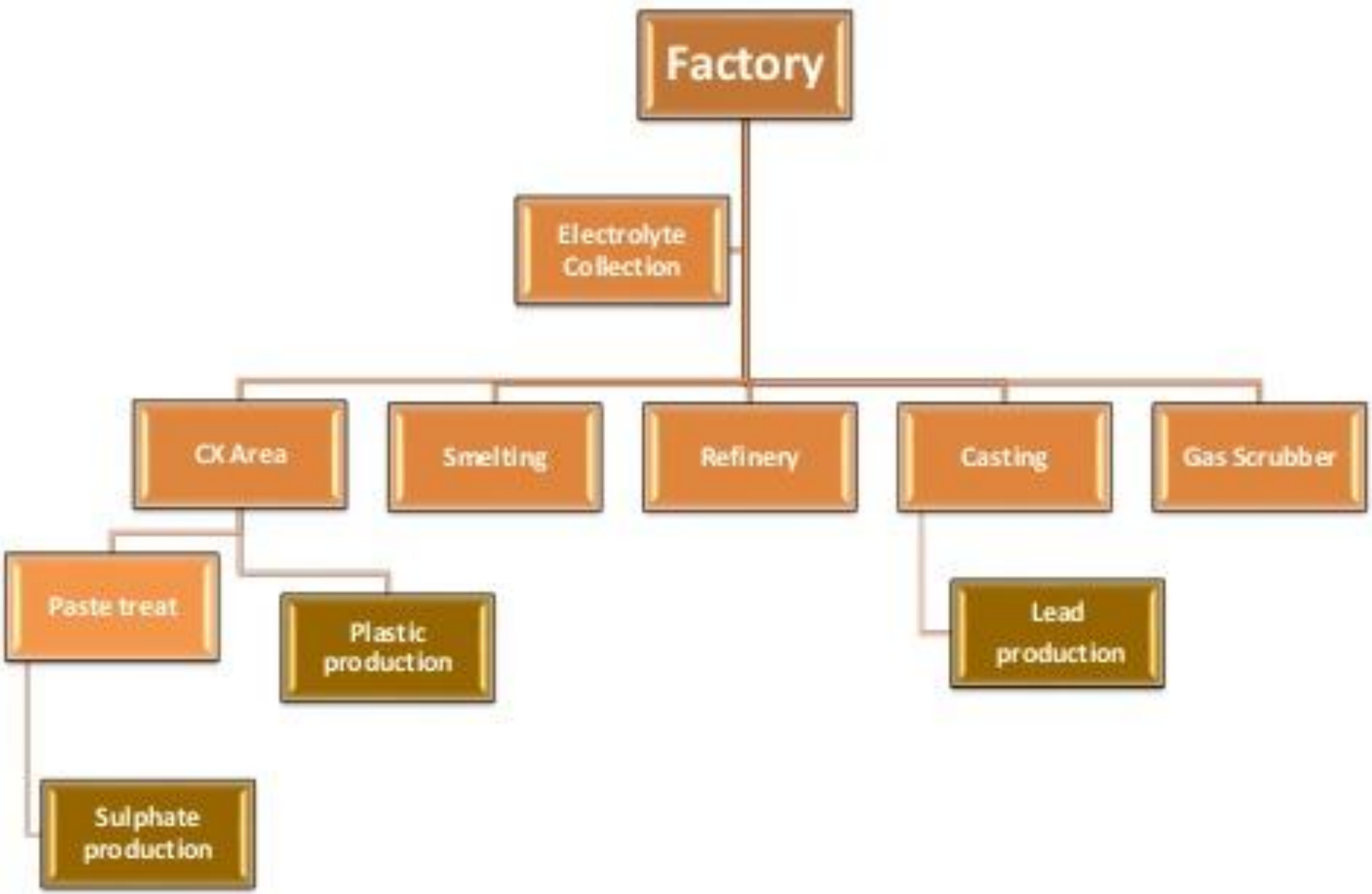
Nickel metal hydride Recycling



Environment monitoring

- Control measures ensure that operational mistakes and accidents are decreased as much as possible,
- In a modern lead recycling plant, the cost of pollution treatment, including effluents, smoke and dusts removal, and elimination of sulphur dioxide (SO_2), amounts to **20-30% of the investment costs.**

Safety in processing



Battery safety

Current flow in the human body

Body Resistance	Current (mA @ 120 volts)	Description
3000 ohms	42 mA	Larger Person
1750 ohms	72 mA	Average Person
1230 ohms	102 mA	Petite Person

- When the human body acts as a "load" it has resistance. The larger the person the more resistance they carry.
- Follow "OHMS" Law to determine actual current at different voltages.
- Keep working conditions DRY, including sweat (ohms law)
- Hybrid vehicles achieve voltage levels up to 650 Volts! That's over five times as high as the above examples.

PPEs

	Locations									Others
1	CX area	√	√	√	√	-----	√	√	-----	-----
	Batteries Bunker	√	√	√	√	In case of handling Soda And Acid	√	√	In case of handling Soda And Acid	
	breaker	√	√	√	√		√	√		-----
	Sulphate process	√	√	√	√		√	√		-----
2	Inside factory gate	√	√			-----	-----	-----	-----	-----

Battery Maintenance and Recycling

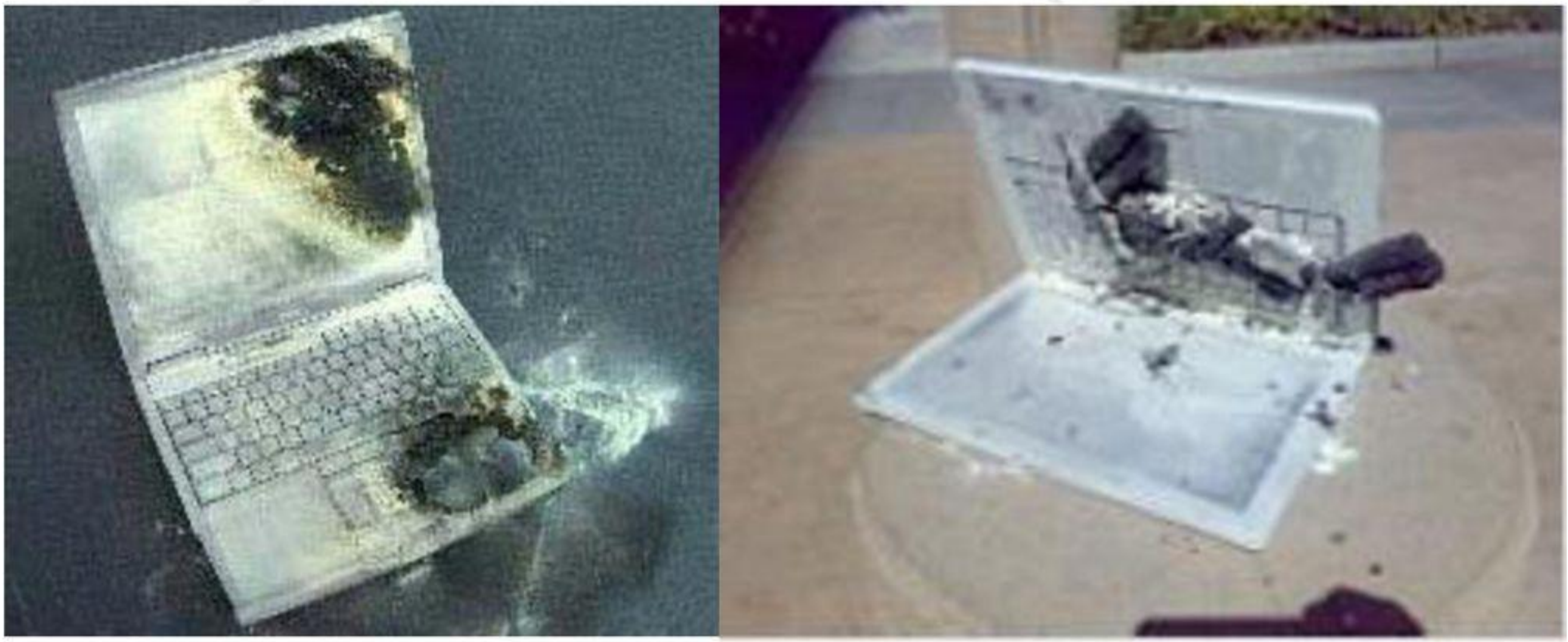
Lead acid battery

https://www.youtube.com/watch?v=eO-X8Gw2nXY

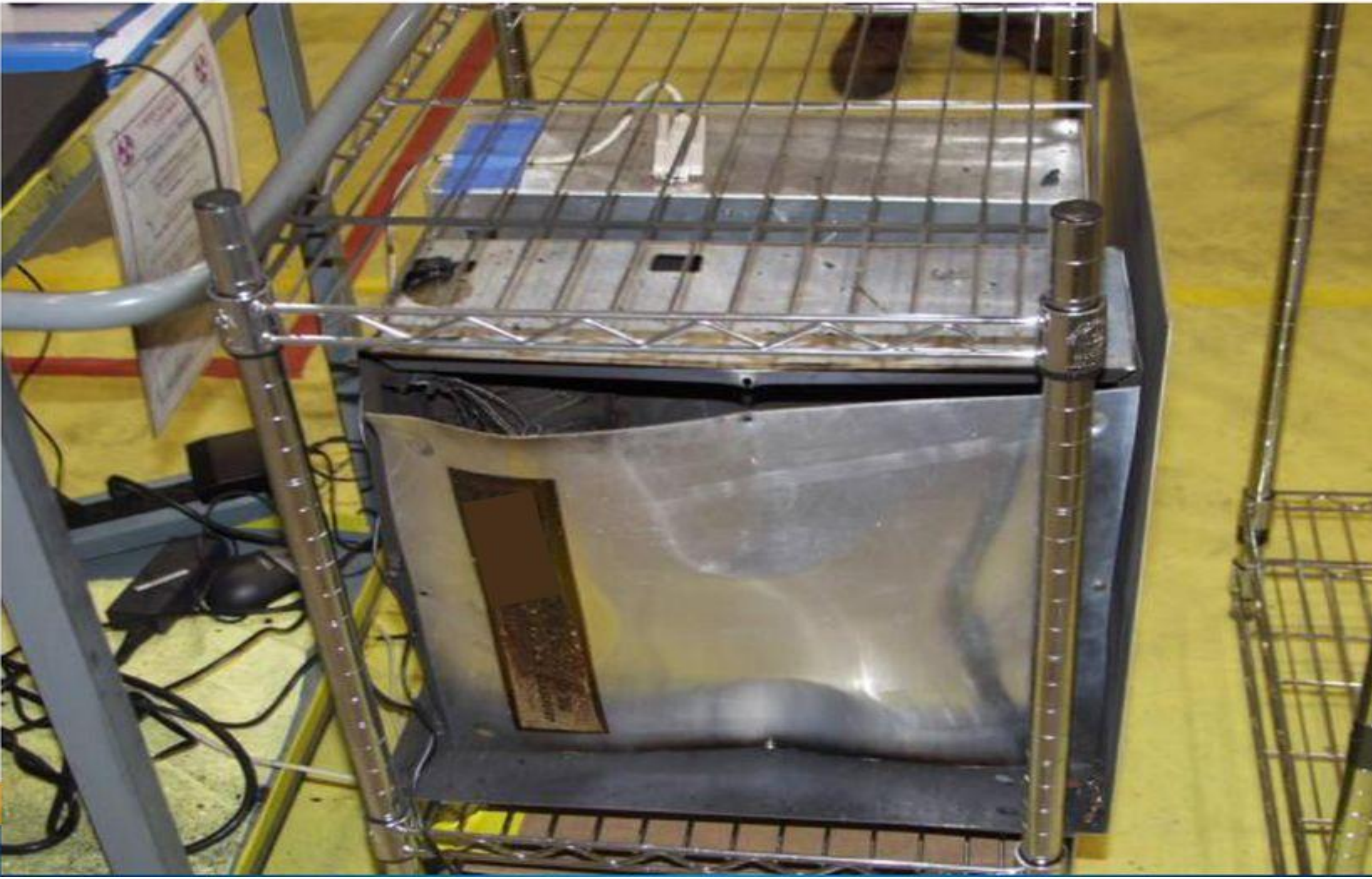
Battery recycling
https://www.youtube.com/watch?v=6w78-aSTIDY

Eco friendly
https://www.youtube.com/watch?v=wxCFDWMPu38

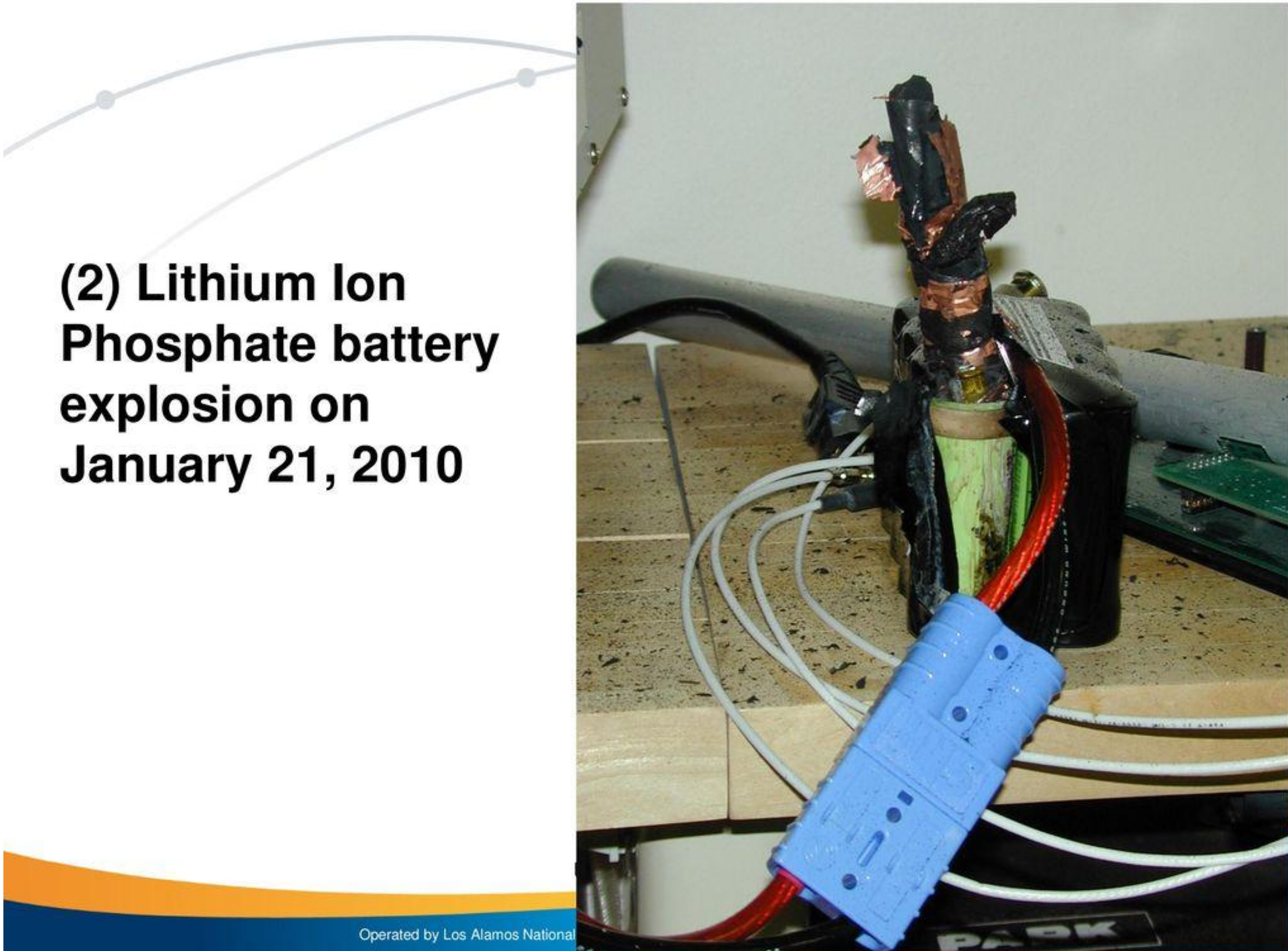
Lithium battery safety and regulations



Lithium battery safety and regulations



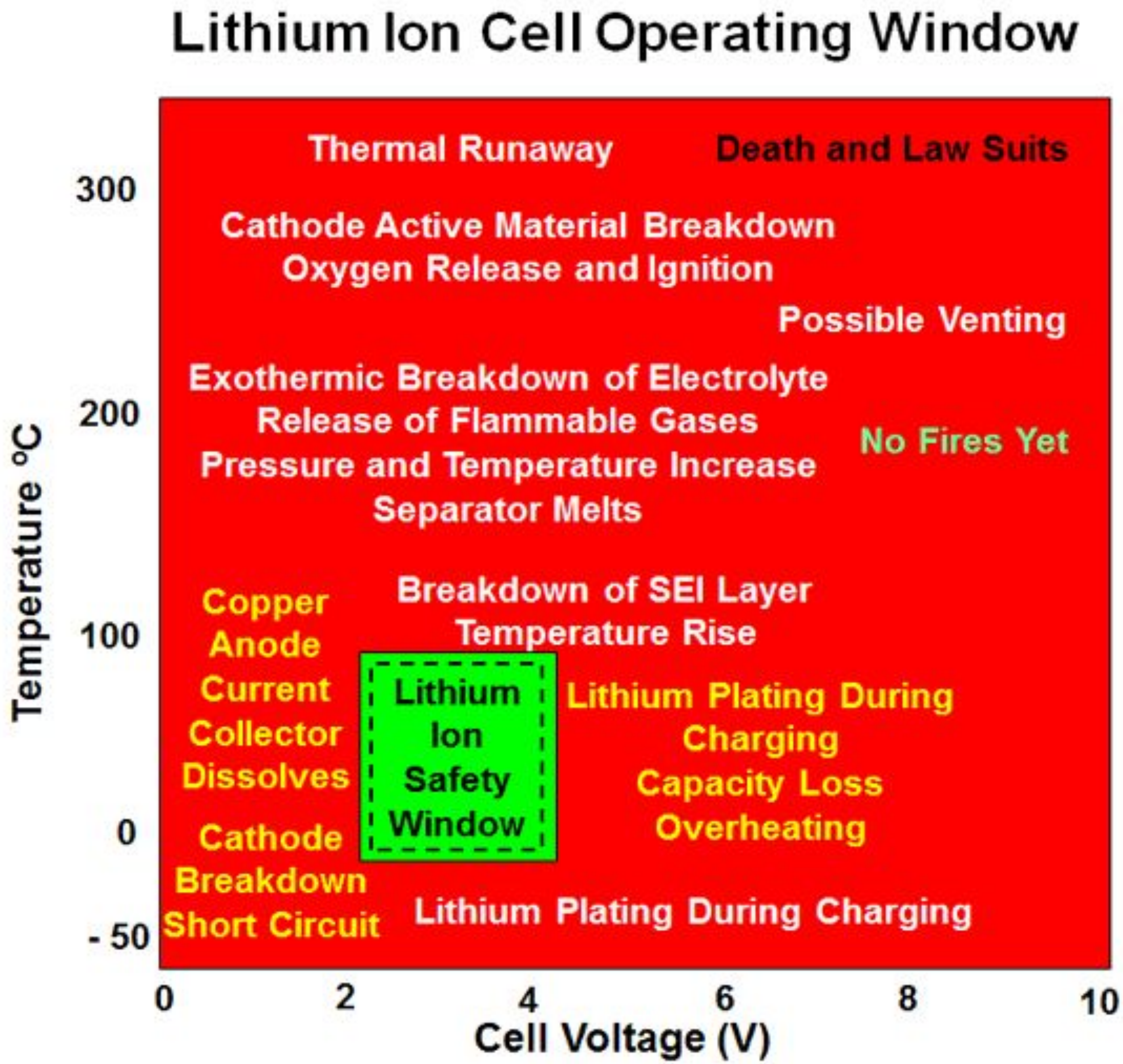
Lithium battery safety and regulations



Lithium battery safety and regulations

- Lithium ion battery fires have recently received additional press due to new products that have caused many reported injuries and fires, including hover boards.
- http://www.bbc.com/news/technology-36727652

Lithium battery safety and regulations



Protection of Li ion batteries

- To prevent hazardous failure, and to maintain safe use, the following protections must be employed
  - protection against overcharge ($V < 4.3$ V)
  - protection against undercharge ($V > 2.5$ V)
  - protection against over charging (less than 1 C in)
  - protection against over discharging (less than 1 C out)
  - protection against over temperature
  - protection against over pressure
- Each cell in a battery pack must be protected
- Series configurations are the most susceptible to failure

Lithium battery safety and regulations

Regulations for Lithium Ion Batteries

- Amazon complies with the following regulations when shipping lithium batteries:
- U.S. Department of Transportation
- United States Postal Service
- IATA's Dangerous Goods Regulations (DGR)
- IATA's Lithium Batteries Guidance
- ICAO's Technical Instructions for the Safe Transport of Dangerous Goods by Air

Lithium battery safety and regulations

- Federal Aviation Agency (FAA) regulations:
  - Spare lithium ion and lithium metal batteries must be carried in carry-on baggage
  - 2 grams of lithium per battery and 100 Wh limit
  - Batteries for further sale or distribution are prohibited
  - With airline approval up to 2 larger lithium ion batteries (101-160 Wh) may be carried on board.

UNCLASSIFIED

Lithium battery safety and regulations

- According to USA today more than 30 universities now ban hover boards on campus due to fire hazard, including:
  - University of New Mexico
  - Purdue University
  - Vanderbilt University
  - University of Massachusetts Amherst

Recommendation in design

1) Li-ion batteries (individual cells) and battery packs (multiple cells connected together) should be properly protected for
   1) under voltage
   2) over voltage
   3) excess charging rate
   4) excess discharging rate
   5) over temperature
   6) over pressure
2) Design, installation, protection, maintenance, and use of Li-ion batteries in equipment should be well-documented, and a rigorous design review should be conducted prior to release for use.

General safety in Li-ion battery

- Store batteries at room temperature
- Do not expose batteries to high temperatures such as could be achieved in a parked car in the sun
- Use a charger designed for the battery or device
- When feasible, do not leave a battery charging unattended
- Store batteries separately from anything hazardous, such as explosives, combustibles, or any other highly flammable material
- Avoid over-charging, over discharging, or shorting
- Watch for signs of failure such as discoloration or swelling
- Dispose of the battery before the expiration date or when the battery is no longer functioning properly

Fire safety in lithium ion battery

- For Lithium (primary, non- rechargeable) batteries: Use Class D extinguishing agent with copper powder. DO NOT use water
- For Lithium-Ion (secondary, rechargeable) batteries: Use an ABC dry chemical fire extinguisher or water hose

In case of contact with electrolyte, gases, or combustion byproducts from a lithium battery or lithium ion battery release:

- Immediately flush eyes with a direct stream of water for at least 15 minutes and seek immediate attention
- Flush skin with cool water or shower for at least 15 minutes and seek medical attention if necessary
- Move to fresh air and monitor breathing and circulation. Take appropriate first aid or CPR actions as necessary and seek immediate medical attention

Lithium battery disposal

If you anticipated generating batteries for recycling, contact your Waste Management Coordinator

- Ensure there is no radioactive or chemical contamination
- Remove batteries from consumer products
- Batteries and battery packs may be disassembled to individual batteries or cells, as long as the battery cell remains intact and closed
- Store batteries that have become waste in a universal waste area
- Label the container with the words “Universal Waste Battery” and the date that the battery became waste

BEST STORAGE AND USE PRACTICES

Procurement

- Purchase batteries from a reputable manufacturer or supplier.
- Avoid batteries shipped without protective packaging (i.e., hard plastic or equal).
- Inspect batteries upon receipt and safely dispose of damaged batteries.

Storage

Store batteries away from combustible materials.

Remove batteries from the device for long-term storage.

Store the batteries at temperatures between 5°C and 20°C (41°F and 68°F).

Separate fresh and depleted cells (or keep a log).

If practical, store batteries in a metal storage cabinets.

Avoid bulk-storage in non-laboratory areas such as offices.

Visually inspect battery storage areas at least weekly.

Charge batteries in storage to approximately 50% of capacity at least once every six months.

Battery Maintenance and Recycling

Chargers and Charging Practice

Never charge a primary (disposable lithium or alkaline) battery; store one-time use batteries separately.

Charge or discharge the battery to approximately 50% of capacity before long-term storage.

Use chargers or charging methods designed to safely charge cells or battery packs at the specified parameters.

Disconnect batteries immediately if, during operation or charging, they emit an unusual smell, develop heat, change shape/geometry, or behave abnormally. Dispose of the batteries.

Remove cells and pack from chargers promptly after charging is complete. Don't use the charger as a storage location.

Charge and store batteries in a fire-retardant container like a high quality Lipo Sack when practical.

Do not parallel charge batteries of varying age and charge status; chargers cannot monitor the current of individual cells and initial voltage balancing can lead to high amperage, battery damage, and heat generation. Check voltage before parallel charging: all batteries should be within 0.5 Volts of each other.

Do not overcharge (greater than 4.2V for most batteries) or over-discharge (below 3V) batteries.

Battery Maintenance and Recycling

- **Handling and Use**
- Handle batteries and or battery-powered devices cautiously to not damage the battery casing or connections.
- Keep batteries from contacting conductive materials, water, seawater, strong oxidizers and strong acids.
- Do not place batteries in direct sunlight, on hot surfaces or in hot locations. Inspect batteries for signs of damage before use. Never use and promptly dispose of damaged or puffy batteries.
- Keep all flammable materials away from operating area.
- Allow time for cooling before charging a battery that is still warm from usage and using a battery that is still warm from charging.
- Consider cell casing construction (soft with vents) and protective shielding for battery research and experimental or evolving application and use.

Battery Maintenance and Recycling